COMPUTER SOCIETY
KNOWLEDGE IS POWER
UNIVERSITY OF JAFFNA

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KANANIYAM

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Digital Twins Explained

Math Matters in the AI Era

THE FUTURE OF Agent Coding

Is Privacy Still a Thing?

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Department of Computer Science, Faculty of Science, University of Jaffna, Sri Lanka

MESSAGE FROM THE EDITORS

It is with great pride and enthusiasm that we present this edition of KANANIYAM. Bringing this magazine to life was a meaningful challenge for us. Within a limited time, we worked with dedication, coordination, and strong teamwork to make this vision a reality.

KANANIYAM reflects the passion, creativity, and academic strength of our computing community. In a rapidly evolving digital world, computer science continues to transform industries, research, and everyday life. Through this publication, we aim to showcase insightful articles on Artificial Intelligence, Data Science, Digital Twins, Smart Technologies, Cybersecurity, and many more emerging areas that shape our future.

We sincerely appreciate the dedication and hard work of all contributors who shared their knowledge and insights. We also extend our gratitude to Dr. S. Mahesan (Visiting Lecturer, DCS) for tirelessly contributing his valuable time to review all the articles and provide comments. His commitment ensures that this magazine remains a platform for knowledge sharing and academic excellence.

We hope this edition inspires readers to think critically, innovate responsibly, and contribute meaningfully to the field of computing. Let us continue learning, exploring, and building a smarter and more inclusive digital world together.

EDITORIAL TEAM

Mr. Raveenthiran Ajan

Mr. Kanesamoorthy Vithursan

Mr. Ravichandran Jananan

MESSAGE FROM THE PRESIDENT

Mr. Ravichandran Jananan
President,
Computer Society,
University of Jaffna.

I am very happy to share this message as the President of CompSoc. During my time as President, we worked hard to improve students' technical skills, career readiness, leadership, creativity, and community involvement.

We organized career guidance programmes, awareness workshops, school outreach workshops, technical workshop series, and special technology initiatives during the year. These included career preparation sessions, AI and cybersecurity workshops, front-end development training, language technology workshops, and a digitization project.

One of the major highlights was UoJCoders v4.0, our flagship coding competition. We also conducted a leadership guest lecture series to guide leadership and strategic thinking.

All these activities were successfully carried out with the dedication of our Executive Committee, student members, and academic staff. Together, we continued to strengthen CompSoc and create a positive impact.

MESSAGE FROM THE PATRON

Dr. (Mrs.) B. Mayurathan

Patron of the Computer Society,
Head of the Department, Department of Computer Science,
University of Jaffna.

With great pleasure, I extend my warm greetings to the Computer Society (CompSoc) of the University of Jaffna as we celebrate another year of dynamic achievements and meaningful engagement.

This year, CompSoc continued its strong commitment to academic growth, career development, and community service. The Society organised several workshops, technical talks, career guidance sessions, and outreach activities for both university students and school students. These programmes helped participants improve their technical knowledge, career readiness, and understanding of new technologies.

The Society's main event, UoJCoders v4.0, once again became a major national-level inter-University coding competition. Teams from universities across Sri Lanka took part in the 12-hour competition, showcasing their programming and problem-solving skills. The successful organisation of this event showed the Society's strong planning and technical skills. These efforts show the hard work, dedication, and leadership of the current committee. As the Patron, I sincerely thank all committee members, volunteers, and academic staff who supported these activities. I am confident that the Computer Society will continue to grow and achieve even greater success in the future.

Wishing the Computer Society continued success in all its future endeavours.

MESSAGE FROM THE SENIOR TREASURER

Dr. Kengatharaiyer Sarveswaran

Senior Treasurer of the Computer Society,
Senior Lecturer, Department of Computer Science,
University of Jaffna.

It is a great pleasure to share a few words for this issue of KANANIYAM, published by the Computer Society.

In today's world, reading as a sustained intellectual practice appears to be gradually declining, with many in the present generation increasingly drawn towards audio-based and multimedia content, often with only the patience for brief social media posts. In such a critical period, the publication of a magazine becomes even more important. It ensures that those who still value reading do not feel left behind and that thoughtful ideas continue to find a meaningful space.

This publication is the result of the tireless effort of the Society's Patron, Dr. (Mrs.) B. Mayurathan, who took the initiative and worked diligently to make this possible despite her many responsibilities.

I would also like to extend my heartfelt congratulations to all the authors who recognise the importance of a magazine even in the AI era and who have generously contributed their time and energy to share their knowledge and insights.

The dedication of the CompSoc committee members and the editorial team is equally praiseworthy. Bringing out a magazine within a short period of time requires commitment, coordination, and perseverance, and their efforts deserve sincere appreciation.

I am delighted that CompSoc continues to uphold this valuable tradition. My best wishes to everyone involved, and I sincerely hope the Society will continue to sustain and strengthen this initiative in the years to come.

MESSAGE FROM THE DEAN

Prof. P. Ravirajan
Senior Professor of Physics,
Dean of the Faculty of Science,
University of Jaffna.

I extend my heartfelt appreciation to the Computer Society (CompSoc) for its outstanding contribution to fostering a vibrant academic and professional community. Your commitment to technological awareness, skill development, and collaborative learning is truly commendable. Through competitions, workshops, hackathons, and knowledge-sharing initiatives, you continue to inspire students to excel in the ever-evolving field of Computer Science. Your outreach programmes for school students further reflect a strong sense of social responsibility and dedication to nurturing future innovators.

I also congratulate the CompSoc on publishing the well-curated 7th volume of the KANANIYAM newsletter. This edition presents a thoughtful and diverse collection of articles that demonstrate both intellectual depth and contemporary relevance. Contributions such as Think Beyond Tools: Why Mathematics Matters in the AI Era and Making Sense of Uncertainty Through Statistics rightly emphasise the foundational role of mathematics and statistics in modern computing. Articles like Is Nature Truly Random? and How the Data Science Process Drives Better Decisions promote analytical thinking and evidence-based reasoning.

The newsletter reflects strong engagement with emerging technologies. Pieces such as Command Centers for Agentic Coding, Digital Twins: Making the Physical World Executable, and Will 2026 Be the Year We Achieve AGI? explore cutting-edge developments in artificial intelligence and autonomous systems. Balanced perspectives are evident in Will AI Replace Software Developers or Empower Them? and When AI Makes Things Up: Understanding Hallucinations, which thoughtfully assess both the opportunities and limitations of AI.

Equally significant are discussions on societal impact and inclusivity. Articles including How Computer Science Shapes Modern Society, The Future of Urban Living Through Connected Technology, The Role of AI in Modern Business Information Systems, and Privacy: Is It Still a Thing Nowadays? highlight the broader implications of digital transformation. Securing Women in the AI Cyberspace underscores the importance of safety and equity, while How Machines Learn the Language of Soil, Sky, and Seeds illustrates interdisciplinary applications in agriculture and sustainability.

Practical insights are well addressed through contributions on AWS Cloud Clubs, version control with Git, and performance testing, effectively bridging academic theory with industry practice.

The dedication and enthusiasm demonstrated by the Society are highly commendable. I look forward to your continued contributions to the advancement of computer science and technology at the University of Jaffna. I extend my sincere appreciation to all contributing authors and to Dr K. Sarveswaran, Senior Treasurer of the society, for his invaluable guidance.

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Digital Twins Making The Physical World Executable

In this era of computing, the digital twin is one of the most evolutionary technologies a concept that tightly integrates the physical world with software. A digital twin is a continuously updated, real-time synchronized virtual model of a real-world object, system or an environment. Digital twinning differs from traditional simulation because it synchronises with real-time data which allows the system to be monitored, analysed, optimised within the virtual environment. This capability makes the physical reality and laws executable within a computational environment [1].

Concept and Evolution of Digital Twins

A digital twin is defined as a dynamic virtual representation that mirrors the structure, state, and behaviour and physical system using continuous data exchange [2]. In smart manufacturing environments, digital twins play a major role by enabling a strong link between physical systems and their digital representations, forming unified cyber–physical systems.

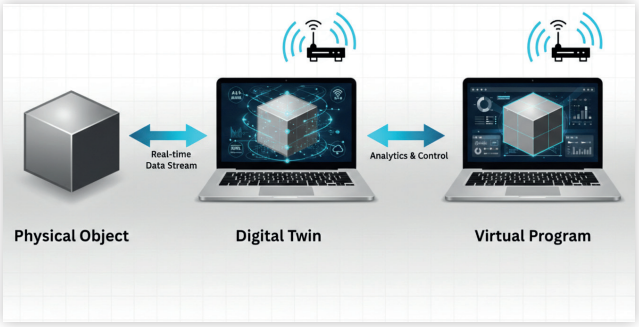
This integration is made possible through enabling technologies such as the Internet of Things (IoT), Artificial Intelligence (AI), and big data analytics, which collectively ensure continuous data flow and real-time synchronization between the physical and virtual environments.

The digital twins differ from the basic modelling and simulation in their ability to support bidirectional, real-time data exchange. This ability allows the digital twin not only to mirror the physical system but also to influence and optimize its operations. As a result, this facilitates real-time analytics, predictive insights, and informed decision-making.

How Digital Twins Work

A digital twin starts with a physical system equipped with IoT sensors that continuously captures data from the physical system. These sensors monitor parameters of the physical system such as temperature, pressure, or motion. Next, this collected data is transmitted through a cloud or edge computing infrastructure for the continuous synchronization between the physical system and its digital representation [3].

The core concepts behind this digital twin are computational models with physics-based simulations, statistical models, or AI driven predictive models. The machine learning algorithms detect patterns, forecast future states and recommend optimum actions by analysing historical and real-time data while the visualisation dashboard enables the user to interact with the system



Digital Twins in Fisheries and Aquaculture

Even though digital twins originated in manufacturing, their applications have spread widely among many domains, and aquaculture and fisheries are rapidly emerging application areas [4]. The digital twins are used in aquaculture to monitor the water quality parameters such as temperature, pH, dissolved oxygen and to model the impact on fish health and growth [4].
Research shows that such systems can predict low oxygen levels, optimise feeding strategies and improve productivity. This technology allows

farmers to act productively rather than reactively by simulating biology and environment integration synchronously, reducing losses and supporting sustainable farming practices.

Why Digital Twins Matter to Computing

Digital twins are integrated with multiple core domains:

- Cyber-physical systems and real time data streaming
- AI and Machine learning
- Simulation and predictive modelling
- Human-computer-interaction and visualization

The integration of these domains makes digital twins an intelligent and powerful framework for next-generation intelligent systems across the industries.

Digital twins show a drastic divergence from the traditional relationship between the software and the physical world. They bridge the gap between computation and reality through real-time monitoring, predictive analytics, and real-time decision-making. Digital twins play a major role in building a sustainable, intelligent technological ecosystem with the advancement of AI, IoT and real-time systems.

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Recent Advances in Computer Science

1. **Multimodal AI**: Modern AI systems can now understand text, images, audio, and video together. Tools like ChatGPT and Gemini can analyze multiple types of input at once, making AI more interactive and human-like.
2. **AI Agents:** Beyond chatbots, new AI "agents" can perform tasks independently such as scheduling, coding, researching, and automating workflows with minimal supervision.
3. **Advances in Quantum Computing:** Companies like IBM and Google are developing more stable quantum processors. Though still experimental, quantum computing could revolutionize cryptography and complex simulations.
4. **Post-Quantum Cryptography:** As quantum machines improve, researchers are building encryption systems resistant to quantum attacks to protect future digital communications.
5. **Cloud-Native & Serverless Architecture:** Modern applications are built using microservices and serverless platforms, allowing automatic scaling and reduced infrastructure management.
6. **AI-Assisted Programming:** AI coding assistants like GitHub Copilot help developers write, debug, and optimize code faster — transforming software engineering practices.
7. **Edge AI:** AI models are now running directly on devices (phones, cameras, IoT devices) instead of relying entirely on cloud servers, improving speed and privacy.

Will 2026 be the Year We Achieve AGI?

Before AGI Where Are We in AI?

Modern Artificial Intelgence(AI) is powerful and excels at imitating humans at specific tasks inseveral areas. Gen AI tools write code, create text, images and videos from a prompt. Digital Assistants around us use Natural Language Processing(NLP) to respond to voice commands.

Automating complex business operations,AI analyses data to calculate, estimate and then suggest decisions. However, the world refers to this as “Narrow AI” as they still execute within carefully defined boundaries. What lies beyond this?

Where Did It All Start?

Artificial General Intelligence (AGI) was not born in a single moment. The idea existed across philosophy, scientific research, and security studies since the late 1990s and early 2000s. Ray Kurzweil's book "The Age of Spiritual Machines" popularized the prediction that by 2030, AI would reach the level of human intelligence. Later, computer scientists such as Ben Goertzel and his colleagues explored how to move AI beyond narrow domains such as playing chess. Through ongoing discussions and research efforts within the AI community, the term AGI was gradually established. However, the term itself first appeared in the paper titled "Nanotechnology and International Security" as early as 1997 by Mark Gubrud.

What Really Is AGI?

There is no universally accepted definition of AGI. Hypothetically it refers to the intelligence of a machine with the human-like ability to understand, learn and apply knowledge to any intellectual task that a human being is capable of. It can be described as mimicking the cognitive abilities of the human brain - which is different from the intelligence behind self-driving cars or medical diagnoses in that demonstrate intelligence only within their specific domains.

Does Turing Test Alone Guarantee AGI?

The “Imitation Game” or the Turing Test is a tool to identify a machines’ ability to exhibit human-like intelligence by conducting a text based conversation with a human judge who tries to distinguish the machine from a real person. In the above process rather than demonstrating intelligence, models may try to simulate human - level errors.

Andrew Ng in one of his newest newsletters mentions that the main goal of AI today should be to build systems that add economical value rather than to fool judges. Therefore, the Turing Test alone is not sufficient to guarantee AGI.

When Will AGI Arrive?

AGI has so far mainly remained a theoretical concept and a research pursuit. It grows with collaboration across interdisciplinary fields such as Computer Science, Neuroscience and Cognitive Psychology. Some researchers suggest AGI may be decades away or may never arrive while others predict that it could arrive in the next few years. These views are reasonable because they are based on different definitions of AGI and varying benchmarks. Whetеher it isbe 2026 or 2045, it is not a prophecy but a statement on when do they think the systems become useful enough to change the world. Yet recently tech giants around the world announce major investments centred on achieving AGI

What Does It Really Matter?

Though this hype marks an important point where AI is shifting from tools to infrastructure. Reliability, Safety, and governance matter most. An intelligent system without aligned objectives could cause harm, even without intention. “Can AI do this?” is not the question anymore but “How do we live with it?”.

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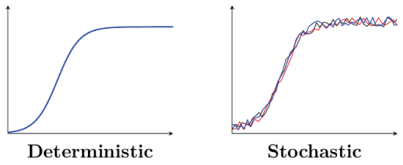
DID YOU KNOW?

- AGI (Artificial General Intelligence) means a machine that can think and learn like a human arross any task.
- Experts say true AGI requires breakthroughs in reasoning, leaming, and adaptability.
- Most predictions suggest AGI is still years or decades away.

Is Nature Truly Random?

We often say nature is random. Leaves grow in irregular directions, seeds scatter unpredictably, and no two trees look identical. Yet variation does not imply disorder. Computer science helps us distinguish between deterministic systems, stochastic processes, and pseudo-random mechanisms, and this distinction clarifies what we mean by uncertainty in nature.

A deterministic system produces the same outcome from the same initial condition. However, determinism does not always mean practical predictability; chaotic systems can be fully rule-based yet highly sensitive to small changes. A stochastic process, by contrast, includes probabilistic elements in its evolution. Individual events may be unpredictable, but the system follows statistical laws. Randomness therefore does not mean absence of structure; it means outcomes are not fully determined in advance, even though probabilities govern them.



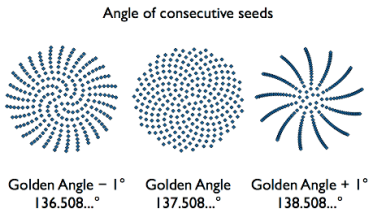
Most natural systems are neither purely deterministic nor purely random. They evolve over time in a state-dependent manner: each new state emerges from the previous one sometimes influenced by stochastic perturbations. In mathematics, this is described as a stochastic dynamical system. The present depends on the past. Natural processes are not independent draws from randomness; they are sequential and structured.

Consider leaf growth in plants. In many species, successive leaves emerge around the stem separated by approximately 137.5°, the golden angle. This arrangement minimizes overlap and improves light exposure. The plant is not "calculating" the angle; rather, local growth rules, hormonal signalling, and physical constraints cause each new leaf to form at the least crowded position.

Over time, this sequential process generates an efficient, non-overlapping structure. Different trees look unique because of environmental variation and biological noise, yet underlying geometric patterns remain visible.

A similar principle appears in reinforcement learning systems in machine learning. When exploration noise is included, the same general starting condition can produce different trajectories, yet these trajectories are not arbitrary; they are generated by structured policies interacting with stochastic elements. Variation arises within rules, not outside them Nature, like well-designed computational systems, balances determinism and randomness; efficiency does not come from chaos, but from structured processes that incorporate controlled variability. The world around us is neither simply random nor rigidly predictable.

It evolves through rule-based dynamics unfolding over time, shaped by constraints and influenced by uncertainty. Between pure determinism and pure randomness lies the rich domain where complex systems grow.



Mr. Y. Pirunthapan
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"Nature is not simply random, nor is it rigidly predictable. Its complexity arises from the interplay of deterministic rules and stochastic variations, creating patterns that are both structured and flexible. Understanding this balance helps us see the order in apparent chaos and the role of controlled randomness in shaping the world around us."

CLOSER THAN CODE:
Understanding Forward Deployed Engineers in Industry

When I started my internship, the term Forward Deployed Engineer was not something I learned from a textbook or a tutorial. I first heard it during our CEO's New Year speech, at a time when the software industry was already changing in visible ways. AI tools were becoming part of everyday development, and writing code-once considered the core of software engineering-was becoming easier with the help ofintelligent assistants.

In that context, our CEO spoke about declaring the comingyearastheYearofProximity.Whatstoodoutwas not a promise of new tools or faster delivery, but a shift in focus. Proximity meant staying close to real problems, real users, and real outcomes. It meant engineers taking responsibility not only for implementation, but also for understanding whether what they built actually worked in practice.

At first, this idea did not feel entirely unfamiliar. As undergraduates, many of us have already worked on projects that involved customers or end users. We have gathered requirements, adapted systems based on feedback, and dealt with constraints that do not appear in assignments. However, we often treated these experiences as academic exercises rather than recognizing their deeper significance.

That significance becomes clearer in the age of AI. Today, AI systems can generate clean, correct, and well-structured code in seconds. Writing code is no longer the hardest part of building software. The harder part is understanding the problem, navigating ambiguity, and making decisions in imperfect environments. This is where the concept of a Forward Deployed Engineer becomes important.

A Forward Deployed Engineer is not defined by a specific programming language or framework. Instead, the role is defined by proximity to the problem space. An FDE works closely with customers, operations teams, and decision-makers to understand how systems behave in real-world conditions. Rather than waiting for fully specified requirements, they help shape solutions by observing workflows, identifying gaps, and translating messy reality into workable systems.

This does not mean abandoning technical depth. Forward Deployed Engineers still write code, design systems, and think critically about architecture. The difference is that their technical work is continuously informed by real usage and feedback. They are involved early, stay engaged after deployment, and take responsibilityforoutcomesratherthanjustdeliverables.

For undergraduates, this perspective reframes experiences we already have. When we interact with users during a project, adjust features based on feedback, or rethink an approach because it does not work in practice, we are already practicing elements of Forward Deployed Engineering. The difference is awareness. Recognizing this role helps us understand why these experiences matter more than ever.

As AI continues to reduce the effort required to produce code, the value of engineers will increasingly lie in judgment, context, and responsibility. Being close to problems, understanding trade-offs, and making informed decisions cannot be fully automated. Forward Deployed Engineering highlights this shift by emphasizing proximity over abstraction.

In this sense, Forward Deployed Engineers are not a replacement for traditional engineers, but an evolution of the role. They represent a way of working that aligns technical skills with real-world impact. For students entering the industry, this mindset can help bridge the gap between academic projects and professional practice.

Understanding Forward Deployed Engineering is not about chasing a new title. It is about recognizing that engineering is ultimately about solving real problems for real people. In an era where code is becoming easier to produce, staying close to reality may be the most valuable skill an engineer can develop.

Ms. Vieronicka, K.
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The DCS successfully concluded its seventh career fair, CS-Career Fair'25 on 29th March 2025, at the Department of Computer Science, Faculty of Science, University of Jaffna.

When AI Makes Things Up

Understanding Hallucinations

Artificial intelligence today can summarize documents, write essays, translate between languages, solve math problems, and even recommend academic research. Because these systems communicate fluently and confidently, users often assume the outputs are reliable. However, many people have encountered a surprising problem: the answer looks correct, sounds authoritative, yet turns out to be wrong.

What Are Hallucinations?

In machine learning research, this phenomenon is called hallucination. It occurs when an AI model produces information that is false, misleading, or unsupported by verifiable evidence,while presenting it with high confidence. This is not intentional deception. Instead, it arises because AI systems are trained to predict likely word sequences based on patterns found in large datasets, rather than verifying facts or checking external sources.

Real-World Examples

- **Academic / Research** - students and researchers sometimes ask AI systems for relevant research papers. The AI responds with well-structured titles, authors, and publication venues. But the papers do not actually exist.
- **Healthcare** - AI may provide incorrect medical advice or treatment recommendations. For instance, it could generate plausible-sounding recommendations that are unsafe or inaccurate, highlighting the importance of verification by qualified professionals.
- **Legal / Policy** - When asked about laws or court cases, AI can confidently produce non-existent legal rulings or misquote legislation, potentially misleading readers
- **News and Journalism** - AI-generated news summaries sometimes include events, statistics, or quotes that never happened, phrased as authentic reportingExample: a startup “raising $100M” when no such funding occurred.
- **Everyday Productivity** - In coding or math tasks, AI might generate incorrect solutions or non-functional code that appears correct because it is convincingly formatted and phrased convincingly.

Consequences and Risks

Hallucinations can cause minor confusion in low-risk situations, but in domains like healthcare, law, journalism, or research, they may contribute to misinformation, flawed decisions, and a loss of trust. Because AI outputs often appear confident,users may trust them without proper verification.

Mitigation Strategies

Reducing hallucinations is an active area of research. Approaches include grounding responses using external data, retrieval-augmented generation, clearer model limitations, and human oversight. However, hallucinations are a structural limitation of probabilistic language models, not just a software bug.

In conclusion, AI can be a powerful assistant, but its outputs should always be verified carefully, especially when accuracy matters. Understanding hallucinations helps users adopt a critical mindset, using AI responsibly while benefiting from its capabilities.

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How the Data Science Process Drives Better Decisions

In recent years, data has become a central asset in decision-making across industries, research domains, and public sectors. Organizations increasingly rely on data-driven approaches to reduce uncertainty and improve the quality of decisions. The data science process provides a structured framework that transforms raw data into actionable insights, enabling informed and evidence-based decision-making. The process begins with problem identification and data acquisition. Clearly defining the objective is essential to ensure that data analysis aligns with decision-making requirements. At this stage, relevant data is collected from multiple sources, including databases, sensors, web platforms, and domain-specific repositories. The relevance, completeness, and reliability of the collected data directly influence the effectiveness of subsequent analysis.

Once data is collected, data preprocessing and exploratory analysis are performed. Real-world data often contains missing values, noise, and inconsistencies that must be addressed through cleaning and transformation techniques. Exploratory Data Analysis (EDA) allows analysts to understand the underlying structure of the data by identifying trends, distributions, and relationships. These early insights help refine assumptions and guide decision-makers before predictive modelling is applied.

The next stage involves data modelling and analytical techniques. Statistical methods and machine learning algorithms are used to extract patterns, predict outcomes, or group similar observations. The choice of model depends on the nature of the problem and the type of decision to be supported. Properly developed models assist organizations in evaluating alternatives, anticipating future scenarios, and minimising risks associated with uncertainty.

Following model development, evaluation and interpretation play a critical role in decision support. Models are assessed using appropriate performance metrics to ensure reliability and validity. Interpreting results in a transparent and understandable manner is essential, particularly when decisions have significant social, economic, or ethical implications. Clear interpretation enhances trust and facilitates communication between technical experts and decision makers.

The final stage is deployment and integration into decision-making systems. Insights are communicated through reports, dashboards, or automated tools that support operational and strategic decisions. Continuous monitoring and model updating processes ensure that decisions remain accurate and relevant as new data becomes available.

In summary, the data science process enables a systematic transition from data collection to informed decision making. By following a structured and iterative approach, data science supports accurate analysis, improves transparency, and enhances the overall quality of decisions. As data continues to grow in scale and complexity, the role of the data science process in driving data-driven decisions becomes increasingly important.

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"Data is a precious thing and will last longer than the systems themselves."
– Tim Berners-Lee

How Machines Learn the language of soil, sky, and seeds

Experience and tradition are no longer sufficient in predictive farming. The field of agriculture is becoming predictive and analytical [2] because of the growth of artificial intelligence intelligence (AI),machine learning, and data-driven computing Computer-based predictive tools help with crop recommendation, harvest forecasting, and environmental effect assessments. Historical crop data, soil information, weather forecasting, satellite imaging and market trends help make a better decision and promote predictive farming.

Predictive Analytics Using Historical Crop Data

Historical crop records build the foundation of predictive farming. Data such as crop type, yield, sowing and harvesting periods, fertilizer usage, and pest or disease occurrences combined with historical weather parameters like rainfall, temperature, and humidity are used to train supervised machine learning models. Before analysis, data undergo preprocessing steps, including cleaning, normalization, and feature extraction. Algorithms like random forests and gradient boosting models learn relationships between environmental conditions and crop outcomes, enabling accurate yield prediction and predicted crop planning [1].

Soil Intelligence and Nutrient Prediction

Soil health is a major factor affecting agriculture productivity [2]. Machine learning models use satellite imagery, IoT sensors and weather data to predict the pH and soil nutrients such as nitrogen (N), phosphorus (P), potassium (K). Spatial and temporal soil variations, reducing dependence on laboratory testing, are captured by techniques like XGBoost and neural networks. These predictions minimize the environmental impact by allowing targeted fertilization and improving resource efficiency.

Weather Forecasting with AI and IoT

Accurate weather forecasting supports irrigation planning, pest control, and crop scheduling. Satellite data, IoT-based weather sensors

and historical climate records are integrated with AI systems to generate localized forecasts. Temporal and spatial patterns of weather data are captured by the models like Long Short Term Memory (LSTM) Networks and CNN-LSTM hybrid. These systems provide early warnings for the extreme weather events and climate resilient farming decisions.

Satellite and Drone Imagery in Precision Agriculture

Remote sensing is an important part in crop monitoring and disease detection [3]. The Normalized Difference Vegetation Index (NDVI) is a vegetation health index that helps in estimating crop health and predicting yield. Satellite imagery provides large-scale coverage; high-resolution drone (UAV) imagery improves accuracy for field-level analysis. When this is combined with deep learning models, these images support precise crop health mapping and targeted interventions.

Market Demand - Driven Crop Planning

Market dynamics are another aspect of predictive agriculture. Historical price trends, seasonal consumption patterns and socio-economic factors analysed by the AI market demand forecasting. Farmers use the market information with crop and environment to make decisions avoiding overproduction and reducing the losses. AI, machine learning, IoT, and remote sensing are shaping agriculture into a smart, data-driven ecosystem. Modern farming enhances productivity and sustainability through accurate forecasting, optimised resource usage, and predictive analytics systems. These technologies hold an immense potential for ensuring food security in the face of climate and economic challenges with their evolving transformations.

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'Decode the Future: CS Terms You Should Know

1. **Artificial Intelligence (AI):** Simulation of human intelligence in machines that can learn, reason, and make decisions.
2. **Machine Learning (ML):** A subset of AI that enables systems to learn from data and improve performance without being explicitly programmed.
3. **Deep Learning:** An advanced type of machine learning that uses neural networks with multiple layers to analyze complex patterns.
4. **Large Language Models (LLMs):** AI models trained on vast amounts of text data to understand and generate human-like language.
5. **Cloud Computing:** Delivery of computing services (servers, storage, databases, software) over the internet.
6. **Edge Computing:** Processing data closer to where it is generated (devices/sensors) instead of relying solely on cloud servers.
7. **Blockchain:** A decentralized digital ledger that records transactions securely and transparently.
8. **Cybersecurity:** The practice of protecting systems, networks, and data from digital attacks.
9. **Internet of Things (IoT):** A network of interconnected physical devices that collect and exchange data.
10. **DevOps:** A culture and set of practices that combine software development and IT operations to deliver applications faster.'

AWS Cloud Clubs:

A Global Student Cloud Community

Amazon Web Services (AWS) Cloud Clubs are student-led communities that help students who are 18 years old and above to learn cloud computing and gain practical experience with AWS. These clubs work inside universities and they mainly focus on knowledge sharing. Every AWS Cloud Club has a Cloud Club Captain, who is a student leader. The captain's job is to develop the local AWS community and connect students with peers around the world [1].

AWS Cloud Clubs are not limited to learning AWS. They also help students become more confident, develop teamwork skills and leadership abilities. These things are really important in university and later when entering the professional industry.

How AWS Cloud Clubs Started

AWS launched the Cloud Clubs programme in 2023. The main goal was to give university students better access to cloud skills. At that time cloud technology was growing very rapidly but most students could not easily access real cloud platforms or perform practical work.

Since then, the programme expanded significantly and now it's in many countries. Today AWS Cloud Clubs are in universities across more than 62 countries with over 500 groups and more than 113,000 members worldwide [3]. This rapid growth demonstrates the programme's success. Now AWS Cloud Clubs is an award-winning programme that helps students learn new skills, build innovative solutions and get ready for their careers.

AWS Cloud Clubs were initially established at a limited number of universities in Sri Lanka. The Institute of Technology University of Moratuwa (ITUM) and the University of Kelaniya started the first cloud clubs in Sri Lanka. More recently, the University of Jaffna and the National Institute of Business Management (NIBM) have received official approval from "AWS Cloud Clubs" and is currently in the process of completing formal university registration. This shows the growing recognition of Sri Lankan universities in the global AWS Cloud Club community.

In Sri Lanka, there is also the "AWS User Group Colombo" where cloud professionals and a community of people interested in cloud computing [4]. Through this group we connect with professionals already working in the industry and learn from what they've done. Having local support like this plus the global AWS Cloud Clubs network really improves learning opportunities.

Leadership Behind AWS Cloud Clubs

The AWS Cloud Club has a good leadership structure that functions effectively. Each university can have only one captain. The captain gives overall vision and direction for the club. They lead students who are interested in cloud technology within the campus and guide them to become AWS experts. Captains organise workshops where students can practice, cloud learning sessions and workshops and technical events that help students understand AWS services in a practical and engaging way.

Captains also get to work with AWS experts and industry professionals which brings valuable learning experiences to club members. Captains manage the club's Meetup page, organise events and participate in the global student Slack community with other captains. This helps them improve their leadership and communication abilities.

To manage cloud club activities, the captain can select two to five members as a "Core Team". Captain can delegate tasks to each core team member. Being in the Core Team provides valuable experience for students interested in leadership roles. One of the Core Team members will become the next captain.

What Students Gain as AWS Cloud Club Members

Members learn by working with cloud technologies that companies use. Through guided projects and organised lessons, students learn about AWS services related to computing, storage, networking, databases and security [1].

Members also gain opportunities to expand their professional network by meeting other students who have similar interests. They can attend sessions with industry professionals, meet AWS experts and receive vouchers for AWS certification exams. Most importantly, members become part of a global student community where members support one another and share knowledge.

Conclusion

AWS Cloud Clubs is a global programme that brings students together with cloud technology and industry professionals. These clubs help students improve their technical skills and also grow as individuals, which is essential for entering future technology careers. I believe AWS Cloud Clubs will continue helping students learn and create innovative solutions all over the world.

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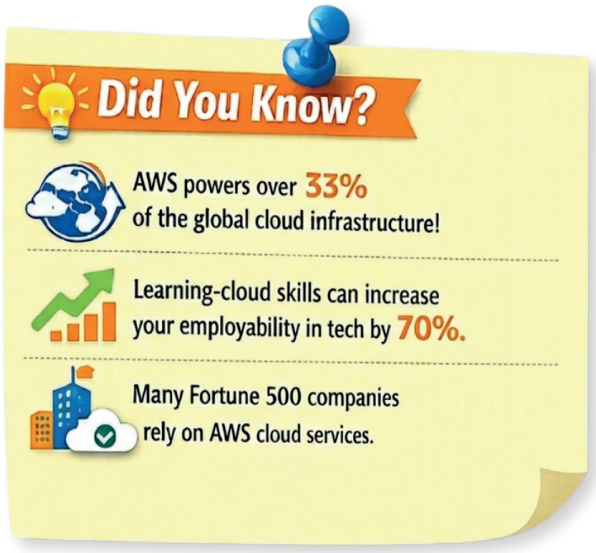

AWS (Amazon Web Services) was officially launched in 2006. The first service AWS introduced was Amazon S3 (Simple Storage Service) in March 2006, followed by Amazon EC2 (Elastic Compute Cloud) in August 2006.

Making Sense of Uncertainty
Through Statistics

Uncertainty is a common part of everyday life. Many decisions are made without knowing the exact outcome in advance. Statistics help us understand and manage this uncertainty by using data and numerical methods to study patterns, relationships, and trends.

Statistics involves collecting data, organizing it in a meaningful way, and analysing it to draw conclusions. Real-world data often shows variation, so results are not always precise. Tools such as tables, graphs, averages, measures of spread, and probability help summarize data and explain uncertain situations clearly.

One familiar example is weather forecasting. Weather predictions are made by analysing past climate data and identifying patterns over time. Although forecasts may not always be accurate, statistics helps estimate the most likely weather conditions. In healthcare, statistics is widely used to study diseases, test new medicines, and compare treatment outcomes, even when patient responses vary. Clinical trials rely on statistical evidence to decide whether a treatment is effective or not.

Statistics also plays an important role in technology and computer science. Many modern tools, such as machine learning models, programming in R, and data analysis applications, depend on statistical methods. These systems process large amounts of data and use probability to make decisions when outcomes are uncertain. For example, recommendation systems suggest products or videos based on past user behaviour and statistical patterns.

In hypothesis testing, statistics provides a clear rule for decision-making using the p-value. A commonly used standard level of significance is 0.05. If the p-value is greater than 0.05, we conclude that the observed result may have happened by chance, and there is not enough evidence to reject the null hypothesis. This method helps researchers avoid incorrect conclusions being drawn and ensures decisions are based on data rather than assumptions.

Statistics is also important in education. For example, Advanced Level (A/L) examination results are analysed using the standard normal distribution. Based on the standard normal distribution, students' performance are converted into Z-scores. The Z-score indicates how far a student's result is from the average and allows fair comparison of results across different subjects and examination years. This method reduces bias and helps maintain fairness in university selection.

In business and economics, statistics helps organizations make informed decisions. Companies use statistical analysis to study customer preferences, predict future demand, and measure performance. Surveys, sales reports, and market research data are analysed statistically to reduce uncertainty and improve planning.

In everyday life, statistics helps people understand exam results, survey findings, news reports, and population data. By analysing data carefully, statistics transforms uncertainty into useful information that supports better judgment.

In conclusion, statistics is essential for making sense of uncertainty. Although uncertainty cannot be completely removed, statistical methods help us analyse data clearly, evaluate evidence logically, and make informed decisions in many areas of life.

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"In a world full of uncertainty, statistics is the compass that guide us to informed choices"

The Future of Urban Living
Through Connected Technology

The rapid growth of urban populations has created major challenges for cities, including traffic congestion, energy shortages, pollution, pressure on healthcare systems, and inefficient resource management.

Traditional city management methods often struggle to handle these demands effectively. As a result, connected technology has emerged as a powerful solution. By using digital networks, smart devices, and real-time data, cities can improve efficiency, sustainability, and the overall quality of life. This technology allows different urban services such as transportation, energy, healthcare, and public safety to work together seamlessly. Connected technology plays a vital role in developing smart infrastructure and intelligent transportation systems. Sensors and monitoring devices in roads, buildings, and utilities help detect problems at an early stage, reduce maintenance costs, and improve safety.

Smart traffic management systems reduce congestion by adjusting signals in real time, while smart parking and GPS-enabled public transport improve convenience while reducing pollution. In addition, smart energy systems such as smart grids and meters help manage electricity efficiently and support the use of renewable energy, making cities more environmentally friendly.

Healthcare, public safety, and environmental management are also greatly improved through connected technology. Remote healthcare services, wearable devices, and telemedicine increase access to medical care and reduce the burden on hospitals.

Smart surveillance, emergency response systems, and disaster warning technologies enhance public safety. Environmental monitoring systems track air quality, water usage, and waste management, helping cities become cleaner and healthier.

Connected technology also provides significant social and economic benefits. Smart city systems improve public services and make urban areas more liveable and inclusive. They create new job opportunities in technology-related fields and support business growth through better infrastructure and connectivity. Digital platforms encourage citizen participation in decision-making, increasing transparency and trust between governments and communities. Although challenges such as data privacy concerns, cybersecurity risks, and high costs remain, effective planning and secure systems can address these concerns, making connected technology essential for the future of urban living.

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"The cities of tomorrow are not built with bricks alone-they are connected, intelligent, and adaptive."

UoJCoders v4.0
National Coding Competition

The Computer Society (CompSoc) of the University of Jaffna, in collaboration with the Department of Computer Science, successfully organized the fourth edition of its national-level coding competition, UoJCoders v4.0. 'UoJCoders v4.0' is a 12-hour inter-university competition aimed at enhancing the problem-solving and programming skills of undergraduate computing students. The event served as a platform for undergraduate computing students across Sri Lanka to showcase their programming and problem-solving abilities while fostering collaboration and innovation among participants.

Registration for the competition opened on 9 July 2025, attracting 95 teams from universities and institutions across Sri Lanka. The online elimination round was conducted via HackerRank on 30 July 2025 to assess participants' coding abilities. Team selection for the finals was based on leaderboard rankings, minimum scoring criteria, and participation, with a strong emphasis on inclusivity and fairness. To provide greater real-world competition exposure and to minimize the impact of potential AI tool misuse during the online phase, a larger pool of teams was advanced, resulting in 74 teams qualifying for the final round.

The final round was held on 23–24 August 2025 at the Department of Computer Science, University of Jaffna, The event commenced with an opening ceremony, where the Vice-Chancellor of the University of Jaffna and the Dean of the Faculty of Science attended the inaugural session and delivered their speeches. The competition ran through three intensive sessions, combining technical challenges with teamwork under pressure, alongside meals and social activities that encouraged interaction and relaxation.

Vice Chancellor, University of Jaffna

Head, Dept.of Computer Science

Dean, Faculty of Science

Out of the 74 qualified teams, 39 teams comprising over 100 participants competed in the finals, representing major universities across the country. Seventeen teams came from the University of Jaffna; seven each from the University of Moratuwa and the University of Peradeniya; three from the Northern University; two from the University of Ruhuna; and one each from the University of Kelaniya, the Sri Lanka Institute of Information Technology (SLIIT), and the University of Vavuniya. The competition spanned three intensive sessions, blending technical challenges with teamwork under pressure, while meals and social activities offered participants time to interact, relax, and build connections.

Based on the final scores, The winners were Team Glory from SLIIT– 1st place (LKR 50,000), Team Data Dynamites from University of Jaffna– 2nd place (LKR 30,000), and Team TeamX from University of Jaffna– 3rd place (LKR 20,000).

Winner
(Team Glory)
SLIIT

1st Runner up
(Data Dynamites)
University of Jaffna

2nd Runner up
(TeamX)
University of Jaffna

The event was made possible through generous sponsorship and support. Apptimus Tech, as Platinum Sponsor, contributed LKR 200,000, enabling event scaling and enhanced participant experience. Unicom SD, as Silver Sponsor, provided LKR 100,000 for logistics and arrangements. Alumni contributor Mrs. Suba Umadevan donated LKR 100,000 to fund the first and second-place prizes and other essential expenses, while SAGO Overseas Education sponsored LKR 20,000 specifically for the third-place prize.

Dr. E. Y. A. Charles, Senior Lecturer, played a crucial role by preparing competition questions and setting up the programming environment using the University's Learning Management System (LMS), ensuring quality and integrity. The competition was coordinated by Dr. (Mrs.) B. Mayurathan, Patron of the society, Dr. K. Sarveswaran, Senior Treasurer, and with the support of all academic and non-academic staff of the Department of Computer Science.

Team Glory
UOJ
CODERS
APPTIMUS TECH
WELCOME

Code, Collaborate, Celebrate: UoJCoders v4.0 – Unforgettable Memories

I share my experience at UoJCoders v4.0, an exceptionally well-organized inter-university coding competition that truly stood out among the many contests I have attended. Unlike most competitions that use common platforms, this event featured a custom-built LMS with strict access controls to prevent AI usage and plagiarism, and it ran smoothly overnight from 6:00 PM to 6:00 AM. The organizers provided nonstop refreshments and even a surprise birthday celebration, making the night memorable. During breaks, we enjoyed activities such as campfires, music, and games, which helped us relax and build friendships. The award ceremony was held immediately after the contest, with certificates and prizes distributed efficiently. The friendly and supportive organizing team ensured everything ran perfectly, showing me how powerful proper planning and teamwork can be in creating an unforgettable and meaningful experience.

Mr. R. Apiram
Winner - Team Glory
SLIIT

Participating in UoJCoders v4.0, organized by the Computer Society, University of Jaffna, was an unforgettable experience that challenged me to think critically, manage time effectively, and collaborate closely with my teammates, Karenthiran Krishnamoorthy and Jeevakaran Somasuntharam. Over the 12-hour competition, we solved real-time algorithmic problems through the Moodle Virtual Programming Lab (VPL), relying on our own knowledge, problem-solving skills, and teamwork under pressure. Competing alongside talented participants from across Sri Lanka inspired us to push our limits, and winning 2nd Runner-Up was a proud moment. Beyond the award, the event strengthened my analytical skills, resilience, and passion for competitive programming, making UoJCoders v4.0 a truly meaningful and memorable learning journey.

Mr. T. Sarujan
2nd Runner-Up - Team X
University of Jaffna

Organizing Committee

Command Centers for Agentic Coding
The Future of
Autonomous Software Development

Modern software development is rapidly evolving with the rise of artificial intelligence. Consider a development team in the IT department planning a new mobile app: a developer defines the functional requirements for a feature, such as a smart task manager, and then allows AI to handle development task. Using agentic AI, AI agents can plan the system architecture, write and test code, fix any bugs, and prepare a ready-to-review pull request by morning.

In the past, developers and testers would have spent hours or even days completing these tasks without he assistance of AI. By leveraging AI, human workload is significantly reduced, and time-consuming processes are accelerated.

This level of autonomy is now becoming practical through agentic AI and multi-agent command centers, where multiple intelligent agents collaborate like human development teams do. This article explores the concept of agentic AI, its evolution, and highlights OpenAI's Codex as the first practical command center for agentic coding.

Introduction

Agentic AI refers to an artificial intelligence system that can make decisions and take actions independently to achieve goals without being instructed at every step. Unlike traditional AI that generates single responses, agentic systems can plan actions, use tools, reflect on results, and adapt over extended periods with minimal supervision.

The major advancement in this field is multi-agent systems (MAS), where multiple AI agents with specialized roles, such as planning, coding, testing, security, and documentation, collaborate simultaneously. This approach mirrors human team workflows, making software development a natural domain for agentic AI adoption.

Command Centers in Agentic Coding

To coordinate multiple AI agents effectively, a command center is essential. The Command Center enables developers to supervise, orchestrate, and guide agents across tasks while maintaining control over outputs.

Developers assign agents goals, monitor their progress to meet those goals, evaluate proposed changes to the tasks they perform, and intervene when required.

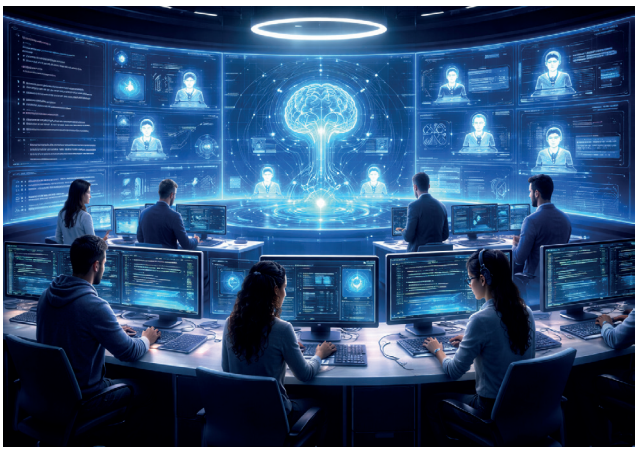
The Command Center automates repetitive and labor-intensive processes so humans can concentrate on design and strategy work and ensure quality. In effect, developers move from being hands-on coders to being conductor or orchestrators of intelligent systems.

Codex Features and Applications

OpenAI's Codex, released on February 2, 2026, exemplifies the agentic AI command center. It lets developers run multiple AI agents in parallel, manage tasks across projects, and review AI-generated changes before applying them.

Codex can automate much of the software development lifecycle, including coding, testing, debugging, documentation, and preparing pull requests, allowing developers to focus on planning, architecture, and strategy.

Codex also includes Skills and Automations to expand its capabilities. Skills allow AI agents to perform tasks beyond coding, such as generating images ormanaging workflows, while Automations run tasks on a schedule in the background. Together, these features make Codex a practical tool for orchestrating AI agent teams and improving productivity.



Challenges

Agentic coding faces several challenges despite its advantages. Long-term reliability is critical, as errors can propagate across multiple AI operations over time. Governance and security are also important, especially when handling private or proprietary code.

High computational costs remain a significant barrier to implementing agentic AI at scale. Additionally, developers and users must provide carefully designed prompts to guide AI agents effectively; unclear or poorly structured instructions can lead to unexpected results or incomplete tasks.

Finally, over-reliance on AI may limit developers' ability to maintain traditional programming skills, as excessive dependence on AI could erode fundamental coding expertise.

Future Outlook

The future of agentic AI and command centers includes standardized agent protocols, multimodal integration, orchestration at larger scales, and stronger verification mechanisms. Developers' roles will increasingly emphasize supervision, governance, evaluation, and human–AI collaboration. Organizations must invest in secure sandboxes, ethical guidelines, and training frameworks to ensure responsible deployment. While Codex currently supports macOS, cross-platform desktop applications for Windows and Linux are expected soon, enabling wider accessibility and adoption.

Conclusion

Command centers for agentic coding represent a major transformation in software development. By enabling intelligent agents to work collaboratively under human supervision, they increase efficiency while preserving accountability. OpenAI's Codex demonstrates the practical potential of this paradigm, marking a shift where developers orchestrate intelligent teams rather than writing every line of code manually. This human-AI partnership signals a new era in computing, where strategic oversight, creativity, and effective orchestration define professional success.

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&
Ms. T. Kayalvizhy
Lecturer,
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Faculty of Computing,
Sabaragamuwa University of Sri Lanka.

Programming IQ Question:

You have the following code snippet:

```
def mystery(x):
    return x * 2 - x // 2
print(mystery(8))
```

What is the output?

A) 10
B) 12
C) 14
D) 16

The Role of AI in Modern Business Information Systems

Business Information Systems (BIS) play a vital role in supporting organisational operations, management, and strategic decision-making. Traditionally, BIS were designed to collect, process, store, and report business data through systems such as Transaction Processing Systems (TPS), Management Information Systems (MIS), and Decision Support Systems (DSS). These systems improved efficiency and accuracy in handling large volumes of data, especially in areas such as accounting, finance, and operations. However, traditional BIS mainly focused on historical data and required significant human interpretation, which limited their ability to support proactive and predictive decision-making.

Artificial Intelligence (AI) has emerged as a powerful technology that significantly enhances the capabilities of modern Business Information Systems. AI refers to the ability of machines to perform tasks that normally require human intelligence, such as learning, reasoning, problem-solving, and decision-making. Unlike traditional systems that follow predefined rules, AI systems learn from data and improve their performance over time. In the context of BIS, AI enables organisations to analyse large and complex datasets, identify patterns, and generate intelligent insights. This transformation allows BIS to move beyond simple data processing and become intelligent systems that actively support business decisions.

AI Technologies in Business Information Systems

Modern Business Information Systems integrate several AI technologies to improve intelligence and efficiency. One of the most important technologies is Machine Learning, which allows systems to learn from historical data and make predictions. Machine learning is widely used in sales forecasting, credit scoring, customer segmentation, and fraud detection. By analysing large datasets, machine learning models can identify patterns that are difficult for humans to detect manually.

Another key technology is Natural Language Processing (NLP), which enables systems to understand and process human language. NLP is commonly used in chatbots, automated customer support systems, and document analysis. In business environments, NLP helps analyse emails, contracts, financial documents, and customer feedback, making it easier to extract useful information from unstructured data.

Robotic Process Automation (RPA) is also widely adopted in modern BIS. RPA automates repetitive and rule-based tasks such as invoice processing, data entry, payroll operations, and bank reconciliation. By automating these tasks, organisations can reduce processing time, minimise human errors, and improve overall productivity. In addition, advanced AI-driven data analytics tools provide real-time dashboards and scenario analyses, helping managers monitor performance and make informed decisions quickly.

How AI Transforms Traditional Business Information Systems

AI transforms traditional Business Information Systems in several fundamental ways. One of the most significant changes is the shift from descriptive analytics to predictive and prescriptive analytics. Traditional BIS mainly focus on explaining what happened in the past, whereas AI-based systems predict future outcomes

and recommend appropriate actions. For example, instead of only reporting last month's financial results, AI systems can forecast future revenue, expenses, and cash flow, enabling better financial planning.

Another major transformation is the shift from manual processing to automation. Many routine business and accounting tasks that previously required human effort are now handled automatically by AI and RPA systems. This reduces the workload of employees and allows them to focus on analytical and strategic activities. Furthermore, AI enables organisations to move from reactive management to proactive decision-making. AI-based BIS continuously monitor business activities in real time and provide early warnings about potential risks, such as fraud or operational inefficiencies. Finally, AI converts raw business data into actionable insights by identifying hidden patterns and generating intelligent recommendations that support managerial decisions.

Benefits and Challenges of AI in BIS

The integration of AI into Business Information Systems offers numerous benefits to organisations. One major advantage is improved decision-making, as AI provides data-driven insights that support accurate and timely decisions. AI also increases efficiency by automating routine tasks, reducing processing time, and minimising human errors. In addition, AI-based BIS enable organisations to gain real-time visibility into business performance, allowing managers to respond quickly to changes in the business environment. These benefits contribute to higher productivity, cost savings, and competitive advantage.

However, the adoption of AI in BIS also presents several challenges. One major issue is data quality, as inaccurate or incomplete data can lead to incorrect predictions and decisions.

Ethical and privacy concerns are also significant, especially when handling sensitive financial and personal information. Organisations must ensure responsible and transparent use of AI technologies. Another challenge is the skill gap, as employees need appropriate training and AI literacy to effectively use intelligent systems. Finally, implementing AI-based BIS requires substantial investment and organisational change, which can be difficult for some organisations to manage.

Artificial Intelligence plays a transformative role in modern Business Information Systems by enhancing their intelligence, automation, and decision-making capabilities. AI enables BIS to move from passive data processing tools to proactive and intelligent systems that support strategic management. While challenges such as data quality, ethics, and skill development must be addressed, the benefits of AI-driven BIS far outweigh the limitations. Organisations that successfully integrate AI into their information systems are better positioned to achieve higher efficiency, stronger competitiveness, and sustainable growth in the digital era.

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"Challenge your mind, unlock the power of AI, turn data into insights, and make every business decision smarter and future-ready!"

Will AI Replace
Software Developers or Empower Them?

In an age where AI is creating transformative changes across the globe, traditional methods of work are changing rapidly. It is already reshaping one of the most sought-after jobs in recent history, that is software engineering. With the burst of large language models (LLMs) like GitHub Copilot and Claude, the lives of software developers have been significantly influenced for software developers. But the question remains as the work of software developers is greatly transformed by the development of AI, will it become a powerful ally for the professionals, or a threat to their careers?

According to a Forbes article in 2025 by Anthropic CEO Dario Amodei, AI will soon write 90% of code, a statement that fuels anxiety within the industry [1]. McKinsey's report suggests that AI in software development can significantly improve quality and efficiency while also preserving careers. Let's explore both sides of the spectrum.

Software Development and AI - The Evolution

AI's involvement in software engineering has made its mark by code auto-completion in IDEs, where most mega-platforms like Amazon's CodeWhisperer and Google's Gemini aid developers in real-time, accelerating development workflows and reducing repetitive effort [4].

History shows us that this fear is not new. With the transition from having low-level programming languages that use bits to the usage of high-level languages in the 1950s, there have been concerns about whether these new advancements might destroy the market. But, instead, it expanded the field of software development, creating more opportunities.

But in 2025, a Medium article suggested that AI-driven productivity gains could replace up to 80% of software engineering roles [2]. But one important point to note is that AI provides quick solutions based on its training models. But when it comes to understanding novel problems, which requires human creativity, AI often struggles.

AI as an Empowerment Tool

AI automates repetitive tasks, significantly reducing development time. Tools like IBM Cognos Analytics make teamwork efficient. Tools like Code Assist help to improve code quality, security, and workflow [3].

Anthropic Internal Use at Anthropic AI transforms software development in a great way. In 2025, they already are using AI for code generation, testing, and reporting to assist their work. The developers found that these make development cycles more efficient and speed up product delivery. But still, human insights are often required to deliver a quality product.

What works: so far the studies showed that bug detection and code optimizations can be areas where AI can be a highly valuable tool. Tools like DeepCode or SonarQube, which use ML integration, can be effective for this. QAT Global studies show that AI enhances code quality and automates testing by showing improvements in work up to 30%. Further, for a medium to small-scale business, AI will cut off the production time by 40%, empowering developers for innovation rather than debugging.

The fear of replacement: myths and realities

As per a Reddit thread from 2025, AI will not replace engineers soon, but such claims often underestimate human value. Meanwhile, Mark Zuckerberg states that Meta envisions AI mid-level engineers writing code, potentially displacing roles.

GitHub's Measurement Approach, where

	Traditional Development	AI-Empowered Development
Code generation	Manual, time intensive	Automated, 50% to 70% faster
Debugging	Human-led, error-prone	AI-assisted, higher accuracy
Productivity	Baseline	Up to 2x gains (per McKinsey)
Innovation Focus	Limited by routine	Enhanced by automation

GitHub with Google and Dropbox measures AI impact through metrics like code acceptance rates. Reports say AI boosts productivity without widespread layoffs. Ascendient Learning's Training Program adopted AI across the SDLC, empowering engineers through hands-on training. Result: Enhanced innovation and reduced context switching, with 77% of developers viewing AI positively.

Yet, junior developers face risks if entry-level tasks vanish as tasks become easier. On X (formerly Twitter), one user commented: "AI didn't replace junior developers. Juniors who only followed tutorials replaced themselves." This highlights adaptation as key. The latest technology may provide novel ways of performing operations, but the sustainability of a professional lies in being flexible and constantly learning to stay ahead of the curve.

As AI is increasingly incorporated into the process of software development, it is essential that ethical issues not be forgotten. Data protection, intellectual property rights, and responsibilities must receive careful consideration within the developer community. It is necessary that AI-produced code is reviewed on a careful and meticulous basis to ensure security and data protection also receive due consideration.

Future Outlook

AI will not outright replace software developers but will profoundly empower them. As per case studies from Anthropic, productivity leaps using AI, where skills such as prompt engineering, system design, critical thinking, and ethical judgment will lead the future roles. The key is balanced adoption, fostering a collaborative ecosystem where technology serves humanity. When used responsibly, AI has the potential to empower developers, accelerate innovation, and create a collaborative ecosystem where technology enhances human creativity rather than replacing it.

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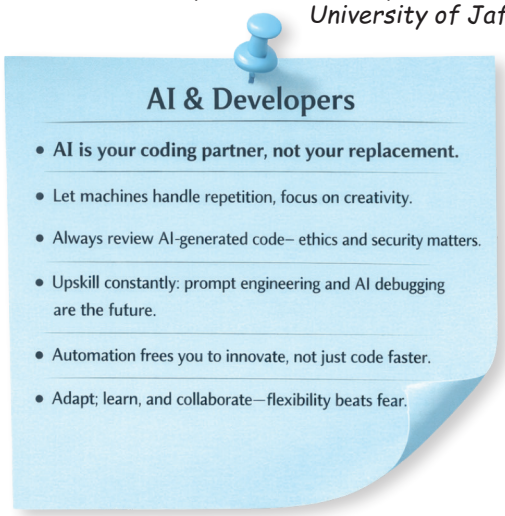


DCS TALENT SHOW 2025

The third-year students of the Department of Computer Science (DCS), with the support of the IEEE Computational Intelligence Society (CIS) Student Branch Chapter, successfully organized the DCS Talent Show 2025 on 27th September 2025, from 9:00 a.m. to 1.30 pm at the Kailasapathy Auditorium, University of Jaffna. The programme provided a vibrant platform for students to showcase their creativity, teamwork, and artistic talents beyond academics. A total of 30 performances were staged, featuring a variety of items such as solo and group songs, dances, instrumental performances, and orchestra pieces. This event was particularly special, as it marked the first-ever talent show organized by the Department of Computer Science.

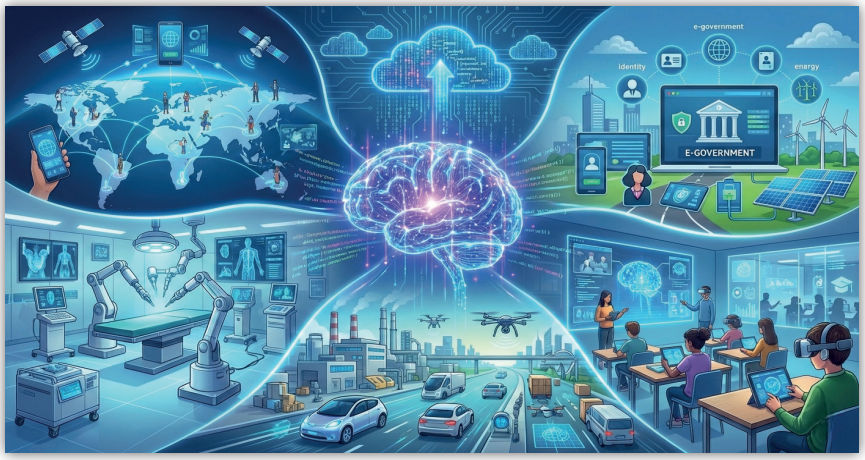
How Computer Science Shapes Modern Society

Computer Science has become one of the most influential disciplines in modern society. Its applications extend far beyond software development and academic research, influencing nearly every sector of society. From communication and education to healthcare, industry, and public administration, Computer Science plays a central role in shaping how modern societies function and evolve.

One of the most significant contributions of Computer Science is in the field of communication. Advances in computing and Internet technologies have transformed the way people share information and interact with one another. Digital communication platforms enable instant exchange of information across geographical boundaries, supporting global collaboration in education, business, and research. As a result, access to information has increased significantly, contributing to knowledge sharing and greater social connectivity.

Computer Science also drives innovation and efficiency within industries. Technologies such as Artificial Intelligence, data analytics, cloud computing, and automation systems allow organizations to improve productivity and decision-making processes. In healthcare, computing systems support medical diagnostics, patient record management, and research. In transportation and manufacturing, automated and automated and intelligent systems enhance safety, accuracy, and operational efficiency. These technological developments contribute directly to economic growth and industrial advancement.

In the education sector, Computer Science has introduced new approaches to teaching and learning. Digital platforms, virtual laboratories, and online learning environments provide flexible and inclusive access to educational resources. In addition, computing education encourages logical reasoning, analytical thinking, and problem-solving abilities, which are essential skills for students in a technology-oriented world.



Public services and governance have also benefited from advancements in Computer Science. E-government systems, digital payment platforms, and data-driven planning improve service delivery and administrative transparency. These systems enable governments to respond more efficiently to the needs of citizens while reducing the time and resource required. Despite its many benefits, the growing influence of Computer Science presents several challenges. Concerns related to data privacy, cybersecurity, ethical use of technology, and unequal access to digital resources must be carefully addressed. Responsible development and use of computing technologies are necessary to ensure that technological progress benefits society as a whole.

In conclusion, Computer Science plays a vital role in shaping modern society by driving nnovation, enhancing communication, and supporting economic and social development. As technological advancement continues, the responsible application of Computer Science will be essential in promoting sustainable and inclusive progress.

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"Computer Science is not just about machines and code — it is the invisible force shaping how we learn, connect, innovate, and build the

Study Smarter:
Free Digital Tools Every Student Should Knows

Category	Tool	Use For
Study & AI Assistant	ChatGPT	Concept explanations, summaries, brainstorming
Study & AI Assistant	Perplexity AI	Research with cited answers
Study & AI Assistant	Khan Academy	Structured lessons in math, science, and computing
Study & AI Assistant	Quizlet	Flashcards and revision practice
Study & AI Assistant	Brainly	Homework questions and peer help
Mathematics	Desmos	Graph plotting and function visualization
Mathematics	GeoGebra	Geometry, algebra, and calculus visualization
Mathematics	Microsoft Math Solver	Solving math problems with step explanations
Mathematics	Photomath	Camera-based math problem solving
Mathematics	Cymath	Algebra and calculus equation solving
Research	Google Scholar	Searching academic research papers
Research	Semantic Scholar	Research discovery and citation tracking
Research	Zotero	Reference and citation management
Research	Mendeley	Organizing research papers and PDFs
Research	Research Rabbit	Visual mapping of related research
Writing	Grammarly	Grammar checking and writing improvement
Writing	Notion	Note-taking and academic organization
Writing	Obsidian	Personal knowledge management
Writing	Microsoft OneNote	Digital lecture notes and documentation
Coding	GitHub Student Developer Pack	Access to student developer tools
Coding	Replit	Online coding environment
Coding	Stack Overflow	Programming problem troubleshooting
Data Science	Google Colab	Python notebooks and machine learning experiments
Data Science	Kaggle	Datasets and data science practice
Data Science	Tableau Public	Data visualization and dashboards
Productivity	Trello	Project and group task management
Productivity	Todoist	Assignment and deadline tracking
Productivity	Google Calendar	Scheduling and reminders
Productivity	Google Keep	Quick notes and reminders
Productivity	Forest	Focus and time management
Presentation & Design	Canva	Creating slides, posters, and visuals
Engineering & Science	Phyphox	Physics data collection using mobile sensors
Engineering & Science	Algodoo	Physics simulations
Qualitative Research	Taguette	Qualitative text analysis

Ensuring a Flawless User Experience with Performance Testing from Development to Deployment

Imagine a situation where you have spent months building an app, and finally, the day of the big release arrives. You expect thousands of users to sign up and start using it. But within minutes, you notice that things start going wrong. For example, pages take too long to load, some features stop working, and, worst of all, the app crashes completely. Users quickly become frustrated and leave, leaving negative reviews and complaints. This is where performance testing steps in. It helps identify problems before they occur, ensuring your app can handle high traffic efficiently. By testing how fast, stable, and reliable your app is under different conditions, you can fix issues early in development and make sure your users have a great experience.

What is Performance Testing?

Performance testing is the process of evaluating how a system behaves under varying loads. It helps answer important questions like:

- Does the system have an appropriate response time?
- Can it handle a large number of users without crashing?
- Will it work well as more users join over time?

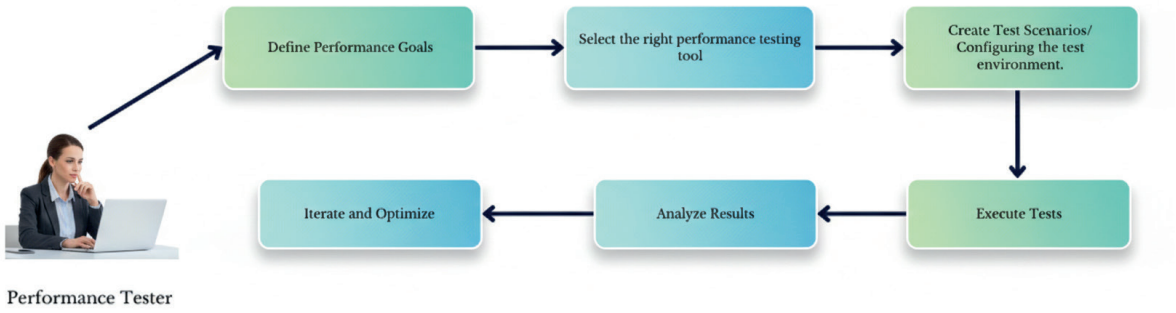
This is different from functional testing, which verifies whether individual features work as expected, like logging in or making a payment. Performance testing, on the other handfocuses on overall system behaviour ensuring the system remains smooth, responsive, and reliable even under heavy use. Performance testing is crucial before deployment because it uncovers hidden bottlenecks and issues that smaller-scale testing may miss. By simulating real-world usage, it reveals problems like slow response times, potential crashes, and scalability limitations. Furthermore, it ensures the application can handle peak loads and remain stable over extended periods of time, preventing issues like memory leaks or performance degradation. Ultimately, performance testing helps validate if the application meets user expectations for speed and responsiveness, leading to higher satisfaction and adoption. Beyond identifying technical issues, performance testing also mitigates risks and reduces costs. Fixing problems before release is significantly cheaper and easier than addressing them in a live production environment. It helps avoid costly downtime and disruptions that can impact business operations and reputation.

Types of Performance Testing

Performance testing includes several types, each designed to evaluate different aspects of a system's performance.

1. **Load testing:** Checks how the system performs under expected user traffic. For example, an e-commerce website expecting 10,000 visitors per day runs a load test to ensure pages load within 2 seconds and checkout remains smooth during peak hours.
2. **Stress testing:** pushes the system beyond its normal limits to determine its breaking point.
3. **Breakpoint testing:** Gradually increases load to determine the exact point where performance starts to degrade.
4. **Soak testing:** Evaluates the system's stability over long periods of operation
5. **Scalability testing:** Measures whether the system can handle increased loads by adding resources like CPU, memory, or additional servers.

Process of Performance Testing



Performance Testing Tools

To ensure smooth and efficient performance testing, various tools are available, each with unique strengths suited for different needs.

Apache JMeter

JMeter is one of the most widely used open-source tools, which means it is free, that supports comprehensive load testing and integrates well with various protocols.

It's ideal for simulating heavy user traffic conditions and analyzing system performance. The scripting is done in XML format through the UI, but you can also use Groovy or Java for advanced features.

JMeter supports many types of protocols like HTTP, HTTPS, FTP, JDBC (for databases), SOAP, REST APIs, email protocols like SMTP, and many others. You can run tests through its GUI or from the command line.

K6

It is a modern, scriptable tool designed for developers, making it a great choice for integrating performance tests into CI/CD pipelines

It provides a developer-friendly JavaScript-based scripting model and provides real-time test execution metrics, making it suitable for automated performance monitoring.

k6 only works through the command line, it doesn't have a graphical user interface.

It supports HTTP, HTTPS, and WebSocket out of the box, and you can test gRPC and other protocols using extensions.

Locust

It is a Python-based open source performance testing tool that allows users to write simple but scalable load test scripts.

It is licensed under the MIT License, which is one of the most flexible and developer-friendly licenses.

Locust gives you a clean web-based user interface to run your tests, control the number of users, and see real-time results. You can also run tests from the command line if you prefer automation.

It has built-in support for mainly HTTP and HTTPS protocols but can be extended to test almost any system.

Conclusion

Finally, to ensure effective performance testing and prevent costly failures, it is important to follow best practices that enhance accuracy and reliability. testing early and often is essential because waiting until production to identify performance bottlenecks can lead to downtime, frustrated users, and revenue loss. Integrating performance testing throughout the development lifecycle helps detect issues before they escalate. Teams must refine system architecture, improve response times and optimise resource allocation to ensure a smooth and efficient user experience. Following these best practices enables organizations to create scalable, high-performing applications that can withstand even the most demanding workloads.

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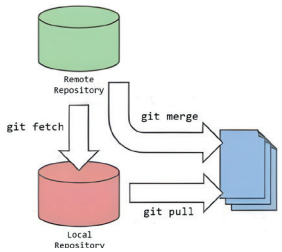
Understanding Git Fetch and Git Pull in Version Control

Version control has become an essential tool in modern software development. Among the many tools available, Git is a free and open-source distributed version control system designed to handle everything from small to very large projects with speed and efficiency. Two commonly used commands in Git are git fetch and git pull. Although they appear similar, their behaviour differs in important ways.

git fetch & pull

pull:
git pull
etch & rebase:
git fetch
git rebase



Git Fetch

```
$ git fetch origin
```

The git fetch command only downloads new data from a remote repository that do not exist in your local repository. However, it does not merge them with your current branch. Fetch is great for getting a fresh view on all the changes that happened in a remote repository.

In simple terms:

- It downloads new commits, tags, and updates from the remote branch.
- It updates your local repository.
- It does not modify your working files or current branch.

Because of this, git fetch is safer. It allows developers to review changes made by others before merging them.

This command is especially useful when you want to see what changes have been made to the remote repository without affecting your local development.

Git Pull

```
$ git pull origin master
```

The git pull command, in contrast, is used with a different purpose in mind: to update your current branch with the latest changes from the remote repository.

Technically git pull is the combination of git fetch and git merge. It will fetch from the remote repository to the local repository and automatically merge them into your current branch.

This leads to two important points:

- Your working files are updated immediately.
- Merge conflicts can occur if your local changes conflict with remote changes.

For this reason, it is strongly recommended to run git pull only when:

- Your working directory is clean.
- You have committed or saved all local changes.

Although git fetch and git pull are closely related, they serve different purposes. git fetch prioritises control and safety, while git pull emphasizes convenience and speed.

A clear understanding of these commands helps students and developers collaborate efficiently, avoid conflicts, and maintain a stable codebase. Mastering version control practices is an important step toward professional software development.

Some Basic Git Commands

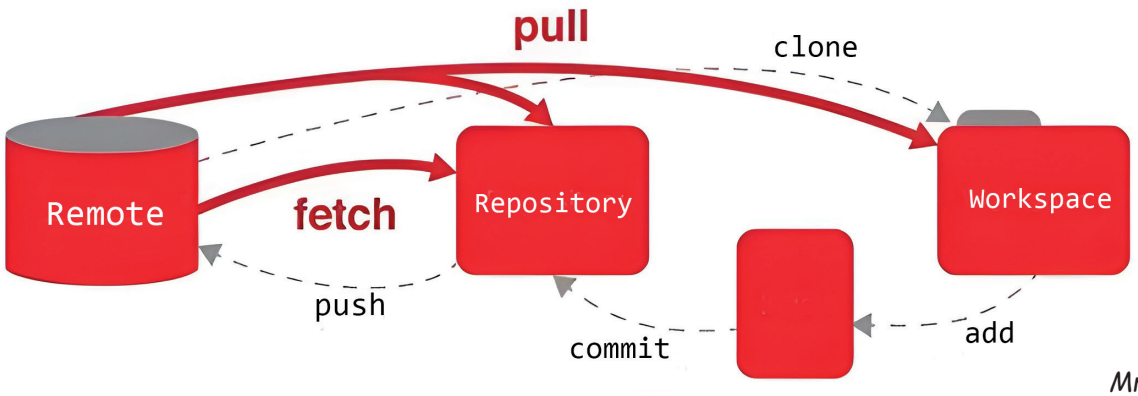
- **git init :** Initializes a new Git repository in the current directory.
- **git clone <url> :** Creates a local copy of a remote repository.
- **git status :** Shows the current status of files and changes.
- **git add <file> :** Adds modified files to the staging area.

- **git commit -m "msg" :** Records the staged changes with a commit message.
- **git log :** Displays the commit history of the repository.
- **git branch :** Lists existing branches or creates a new branch.
- **git checkout <branch> :** Switches to the specified branch.
- **git merge <branch> :** Combines changes from another branch into the current branch.
- **git push origin <branch> :** Uploads local commits to the remote repository.

Comparison of Git fetch and Git pull

Git fetch	Git pull
Command: git fetch<remote>	Pull downloads the new data and merges it into your current working copy files.
Fetch does not change working files	Pull updates working files immediately.
Fetch helps developers review changes before merging.	Pull automatically integrates changes into the local repository
git fetch is a simple command	git pull is similar to git fetch followed by git merge
It allows developers to see commits pushed by other developers before integrating them into the local repository.	It allows the developer to bring changes into the local repository and update it
Fetch protects your code from merge conflicts	There is a higher chance of merge conflicts when using git pull
No merging means no conflicts. They usually occur at a later stage.	Merge conflicts can occur with git pull when two people are working on the same piece of code.
It is better to use the git fetch command with the git merge command on a fetched a repository.	It is not an excellent choice to use git pull if you already pulled a repository.

Workflow of Git fetch and Git pull



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Digital Literacy:
AI & Cybersecurity Awareness Workshop

The “AI & Cybersecurity Awareness” program, organized on 3rd April 2025 by the Computer Society (CompSoc) in collaboration with the Career Guidance Unit of the University of Jaffna and the Career Guidance Cell of the Faculty of Arts, aimed to introduce Arts undergraduates to essential concepts in Artificial Intelligence and Cybersecurity. The event featured expert sessions by Prof. M. Siyamalan and Dr. (Ms.) J. Samantha Tharani, along with practical training conducted in the afternoon by DCS students K. Sanjay, R. Jananan, and K. Vithursan. A total of 112 third- and fourth-year students from Media Studies, Linguistics and English, Translation Studies, and other departments participated. The sessions covered AI fundamentals, ethical concerns like deepfakes, hands-on experience with tools such as ChatGPT and Runway ML, and cybersecurity practices including phishing detection and two-step verification.

The participants gave overwhelmingly positive feedback and expressed strong interest in similar sessions in the future. Encouraged by this response, the Computer Society plans to extend these programs to other faculties, including the Faculty of Ramanathan Visual Arts and Performance and the Faculty of Siddha Medicine. The event’s success was made possible with the support and permissions granted by the Head of the Department of Computer Science, Head of the Computer Unit, and the Dean of the Faculty of Arts, to whom sincere acknowledgements are extended.

Work Smarter, Not Harder:
Exploring Generative AI Solutions
Phishing
Malware
DCS

Think Beyond Tools

Why Mathematics Matters in the AI Era

Would you blindly trust a friend who claims to know all the answers but actually guesses based on predictions? Probably not. So why should we blindly trust AI - Artificial Intelligence? Just like a friend, AI makes predictions, but the engine driving those predictions is mathematics and physics. If we want to truly understand, improve, and innovate with AI, we must first understand the math and physics that power it.

Elon Musk and Nvidia CEO Jensen Huang both emphasise that in the AI era, students should prioritise understanding mathematics and physics over just coding. Tools and libraries-Python, TensorFlow, PyTorch-are powerful, but they are only as good as the minds that design and control them.

Introduction

Are you a Computer Science student wondering why mathematics is there in your curriculum? Ever felt exhausted by complex calculations and thought, “I can just use tools”? This is for you. Modern tools are powerful, and many can use them effectively. But the future belongs not just to tool users; it belongs to those who understand and build the systems behind them. Mathematics transforms a user into a system builder.

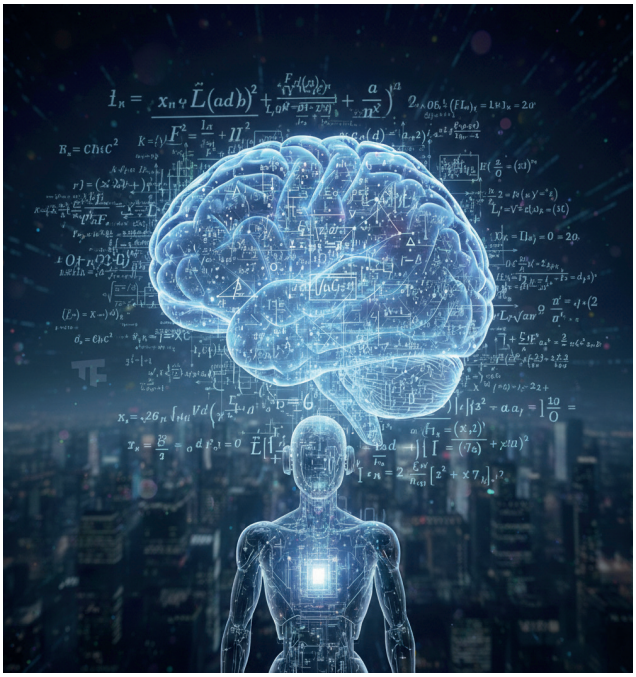
Artificial intelligence (AI) and Machine learning (ML) are transforming industries like healthcare, finance, and entertainment. These systems recognise faces, understand speech, and make complex decisions, but their intelligence is mathematical: data representation, pattern detection, and learning all depend on mathematics. Understanding these foundations is key to moving beyond surface-level usage.

Linear Algebra: The Core of AI

AI data is represented as vectors, matrices, and tensors. Linear algebra enables efficient

AI data is represented as vectors, matrices, and tensors. Linear algebra enables efficient data storage, manipulation, and transformation. Face recognition systems, neural networks, and deep learning rely heavily on matrix operations.

Vector spaces, matrices, eigenvalues, eigenvectors, and the Gram–Schmidt process prepare you for neural networks, optimisation techniques, and dimensionality reduction methods such as Principal Component Analysis (PCA), which are fundamental to modern AI systems.



Probability & Statistics: Decision-Making Under Uncertainty

Real-world data is noisy and uncertain. Probability and statistics allow AI to predict outcomes, detect patterns, and measure reliability. Recommendation systems and medical diagnostics depend on these tools.

Sets, functions, and relations form the basis for probability distributions, feature mapping, and data analysis in AI.

Calculus: Enabling Learning

AI learns by optimising performance. Calculus especially derivatives-drives gradient descent and error minimisation. Autonomous vehicles, robotics, and financial forecasting rely on these principles. Derivatives, optimisation, and linear transformations from your syllabus support gradient computation and model fine-tuning.

Graph Theory: Understanding Connections

Many AI applications involve relationships between data points, and graph theory models these as nodes and edges. Search engines, social media, and recommendation systems rely on graphs to rank and detect patterns. Relations, equivalence classes, and partial orders help model networks and connectivity.

Optimisation: Efficient Solutions

AI often seeks the best solution among many possibilities. Optimisation reduces errors, maximises efficiency, and guides decisions in logistics, autonomous navigation, and business analytics. Linear algebra and quadratic forms allow efficient formulation of objective functions and gradient-based optimisation.

Logic & Discrete Mathematics: Foundations of AI Reasoning

Logic and discrete mathematics enable AI to reason, make decisions, and process symbolic data. Chatbots, decision trees, and automated reasoning rely on these concepts. Proof techniques, Boolean algebra, and discrete mathematics underpin rule-based AI and algorithmic design.

Why Mathematics Matters

As AI advances, mathematics becomes more critical. Modern neural networks, generative models, and quantum AI rely on advanced math. Understanding mathematics allows you to innovate responsibly, reduce bias, and ensure fairness.

Conclusion

Mathematics is not just an academic requirement-it is the driving force behind AI. Data representation, learning, reasoning, and optimisation are all grounded in math. Mastering mathematics transforms you from a tool user into a system builder, capable of shaping the next generation of intelligent technologies.

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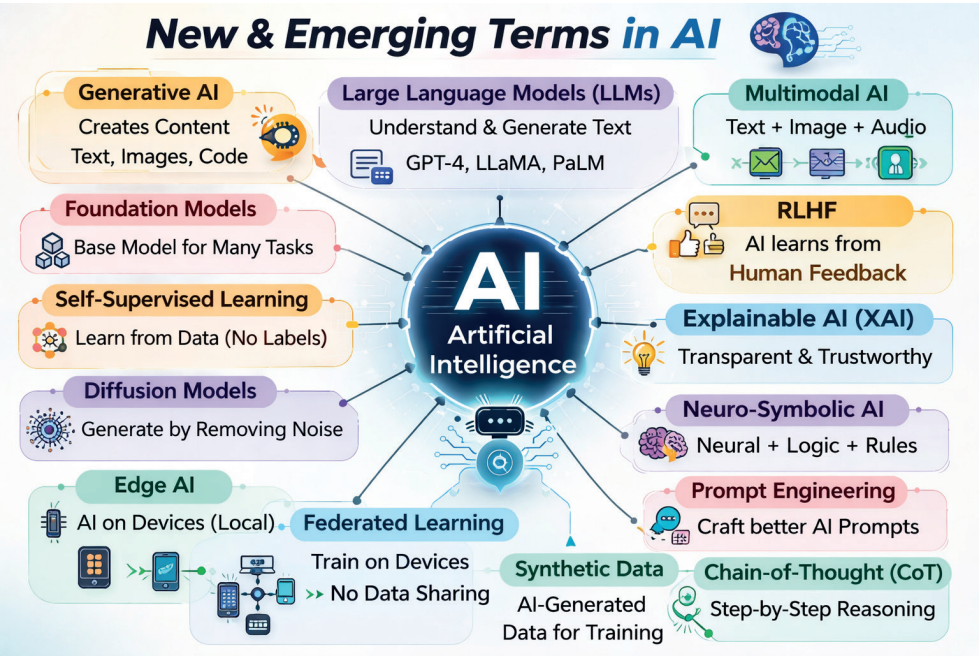

Privacy in Messaging Apps:
Is It Still Possible Today?
Examining WhatsApp and Telegram's End-to-End Encryption Claims

Introduction

In today's digital age, privacy has become one of the most contested issues. With nearly everyone using social media to chat, share updates, and post pictures, the question of how secure our communications truly are has never been more important than today. Messaging apps like WhatsApp and Telegram are among the most trusted platforms worldwide, largely because they advertise strong end-to-end encryption features. But how much of this promise truly holds, and where do the cracks begin to appear?

What WhatsApp and Telegram Advertise About Encryption

WhatsApp's Advertising and Claims

- Your privacy is our priority. With end-to-end encryption on WhatsApp, your personal messages, photos, calls and more stay between you and the people you choose, meaning not even WhatsApp can see them.
- End-to-end encryption: Personal messages, calls, photos, and videos are secured with a lock through end-to-end encryption on WhatsApp, only the recipient and you have the special key needed to unlock and read them.
- We do not store private messages on our servers once we deliver them, and for additional security we provide two-step verification to protect against unauthorised account access. If you lose access to your WhatsApp account, the messages you previously received will remain on your phone and will not be available elsewhere.

Telegram's Advertising and Claims

Telegram presents a different model, highlighting both cloud convenience and optional end-to-end encryption:

- **Cloud Chats:** Telegram describes itself as a cloud service. Messages, photos, videos, and documents from cloud chats are stored on its servers allowing users to access their data from any device without third-party backups. Telegram says all data is "heavily encrypted" and that encryption keys are stored in multiple data centers across different jurisdictions to prevent local engineers or intruders from accessing user data.
- **Secret Chats:** Telegram advertises Secret Chats as its true end-to-end encrypted option. These chats use a key that only the sender and recipient know, meaning "there is no way for us or anyone else without direct access to your device to learn what content is being sent." Telegram adds that Secret Chats are not stored on its servers, no logs are kept, and they are not available in the cloud -they can only be accessed from the devices involved.

How WhatsApp End-to-End Encryption Works

WhatsApp's End-to-end encryption is used when you chat with another person using WhatsApp Messenger. End-to-end encryption keeps your personal messages and calls between you and the person you are communicating with. No one outside of the chat, not even WhatsApp, can read, listen to, or share them. This is because with end-to-end encryption, your messages are secured with a lock, and only the recipient and you have the special key needed to unlock and read them. All of this happens automatically: no need to turn on any special settings to secure your messages.

Where WhatsApp's Claims Break

- **Reporting chats:** When a user reports a conversation, WhatsApp receives up to the last five messages along with metadata about recent calls. This mechanism is designed to

help the company investigate abuse, but it also means WhatsApp can access limited content from private chats. Although WhatsApp claims it is “only five messages,” those five can be drawn from any time period at the moment of reporting. In practice, this means that no message is entirely guaranteed to remain private between sender and receiver once reporting is triggered.

- **Meta AI integration:** Meta is developing “Private Processing” to enable AI features in WhatsApp. Experts warn this could undermine encryption guarantees, since AI systems may require access to message data.
- **Metadata exposure:** Even with encryption, WhatsApp still collects metadata (who you messaged, when, and for how long), which may reveal patterns of communication.

Where Telegram’s Claim Breaks

- **Cloud Chats:** By default, Telegram stores chats, photos, and files on its servers using client-server encryption. This means Telegram itself holds the keys and can technically access your messages.
- **Secret Chats:** True end-to-end encryption is only available in Secret Chats, which must be manually enabled and is limited to one-on-one conversations. Group chats and channels are not end-to-end encrypted. Even in Secret Chats, your account information (such as phone number, name, and IP address) can be disclosed if Telegram receives a valid legal request.
- **Importance:** Telegram claims that Secret Chats are fully encrypted and therefore your data is safe. However, this safety is limited. Because Telegram itself cannot access or analyse the content of Secret Chats, no one can review or investigate if abuse occurs within them. This means that while the messages are technically secure, but limitations remain, users may not be protected from harmful behavior inside these private conversations.

Conclusion

The comparison between WhatsApp and Telegram shows that while both platforms advertise strong encryption, neither delivers

absolute privacy. WhatsApp offers end-to-end encryption by default, but its reporting mechanism, metadata collection, and integration of Meta’s AI features mean that communication is not completely secure.

Telegram, on the other hand, promotes Secret Chats as fully encrypted, yet most regular and group chats remain accessible to its servers, and even in Secret Chats, account information such as phone numbers and IP addresses can be disclosed under legal requests.

The point here is not to decide which app is “better,” but to recognize that no platform provides complete security. What ultimately matters is transparency: these companies should publicly acknowledge their vulnerabilities instead of presenting encryption as flawless. Ultimately, the choice rests with users whether they prioritize data safety (protecting the content of messages) or personal safety (ensuring accountability and protection against abuse). In today’s digital age, privacy is still a thing, but it is never absolute.

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Industrial Visit for

Level 2S Computer Science students – 2022/2023

From Classroom to Industry:
Student Experiences During the Visit

By Department of Computer Science, University of Jaffna - Industrial Visit 2025

Securing Women in the AI Cyberspace

Artificial Intelligence, often called AI, has become one of the most powerful and fascinating technologies of our time. AI helping students learn better in classrooms to supporting doctors in hospitals, is quietly shaping the way we live, study, and communicate. It is designed to make machines think and act intelligenly, similar to humans, and its applications are now found in critical sectors such as education, healthcare, transportation, security, and digital communication. While AI offers many benefits and exciting opportunities, it also raises serious concerns when used in harmful ways.

As AI becomes more advanced, cybercrime has also evolved. In the early days of the internet, cyberattacks were carried out by humans sitting behind screens. These attacks included phishing emails, fake phone calls, online scams, malware, and ransomware, all aimed at stealing personal information or money. Today, with the help of AI tools, cybercriminals can operate faster, smarter, and on a much larger scale. This troubling shift has made women, children, and older adults particularly vulnerable in online spaces.

Recent reports by the United Nations Women (UNW) Organisation highlight the seriousness of this issue. Many women around the world have either experienced online violence themselves or witnessed others being targeted. AI-powered technologies are increasingly being used to harass, threaten, stalk, and blackmail women, causing deep emotional distress and lasting harm. This form of abuse, known as AI-facilitated violence against women, can affect victims psychologically, socially, professionally, and financially, often extending into real-life consequences.

One of the most alarming forms of AI-enabled abuse is image-based violence. AI can now gen commonly known as deepfakes. erate highly realistic fake images and videos, commonly known as deepfakes.Most deepfake content found online involves non-consensual and explicit material, and women are overwhelmingly targeted. These fake images and videos can damage reputations, harm mental health, and leave victims feeling powerless, even though the content is completely fabricated.

Another growing threat is AI-powered impersonation. Cybercriminals use AI to create fake online identities that appear real and convincing. These digital imposters can engage in prolonged conversations, especially on social media and dating platforms, tricking women and girls into sharing private information or images.

This practice often leads to sextortion, where victims are blackmailed and threatened. What makes this particularly dangerous is that AI can imitate human emotions and language so convincingly that it becomes difficult to distinguish between real people and bots.

Artificial Intelligence is also being used to intensify online harassment efforts. Advanced

language-processing tools can analyse social media posts to identify women who speak openly about sensitive issues such as harassment or discrimination. These women are then targeted with personalised threats, abuse, or the exposure of their private information, a practice known as doxing. Such attacks are designed to silence voices and cause fear, leading many women to withdraw from online spaces entirely.

Despite these challenges, there is hope. Around the world, organisations and digital safety initiatives are working to protect victims and make the internet safer. Support services now exist to help remove non-consensual images, prevent private content from being shared, and offer emotional and legal guidance to those affected by online abuse. Learning how to stay safe online and knowing where to seek help are powerful tools, especially for young people growing up in a digital world.

Governments are also beginning to respond. While many countries still lack strong laws to address online abuse, progress continues to be made. New legislation in regions such as the United Kingdom, the European Union, Mexico, and Australia aims to criminalise digital violence and improve accountability for AI-generated harm. However, laws alone are insufficient. Education, awareness, and digital literacy play a crucial role in ensuring that technology is used responsibly and ethically.

For young readers, understanding both the benefits and risks of AI is essential. Technology should be a force for creativity, fairness, and progress not fear. By learning to think critically, protecting personal information, treating others with respect online, and standing up against digital abuse, students can help build a safer and more inclusive digital future.

At the heart of all technological progress lies a simple truth: innovation must serve humanity. When AI is developed and used with care, responsibility, and empathy, it has the potential to uplift society instead of dividing it.

"At the core of successful AI adoption is trust. Innovation that builds trust, is inclusive, and prevents harm is ultimately more sustainable and widely adopted than innovation that undermines those values."

- Paola Gálvez-Callirgos,
expert in AI and digital technology

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The Shift from Chatbots to Digital Workers:

UNDERSTANDING AGENTIC AI

The landscape of artificial intelligence is undergoing a profound transformation, moving beyond the era of impressive but passive chatbots into a new phase defined by action and autonomy. While the initial wave of modern AI introduced the world to systems that could write poetry, debug code, or summarise history, these systems remained fundamentally reactive. They waited for a human prompt and responded with text or images. We are now witnessing the rise of "Agentic AI," a technology that transitions artificial intelligence from a passive oracle of information into an active participant in the digital world. This shift represents a move from systems that merely "think" to systems that can "do," fundamentally changing the relationship between humans and machines from simple interaction to complex delegation.

To truly understand what Agentic AI is, one must look at where it comes from, as the concept of an "agent" is not new to computer science. The dream of autonomous software dates back to the earliest days of cybernetics and the "expert systems" of the mid-20th century. In those early decades, computer scientists attempted to build agents using rigid, rule-based structures. These systems could perform specific actions such as diagnosing a machine failure or routing a phone call but only if the situation perfectly matched the rules programmed into them by humans. They lacked the flexibility to adapt to the messy, unpredictable nature of the real world. If a scenario fell even slightly outside their pre-written script, these early agents would fail. For decades, the "brain" required to navigate ambiguity simply didn't exist, leaving these agents as helpful but limited tools.

The breakthrough that enabled modern agentic AI was the arrival of large Language Models (LLMs). These models provided the missing cognitive layer a probabilistic "brain" capable of understanding context, nuance, and human intent. Agentic AI is essentially the integration of this new cognitive reasoning with the ability to use tools. Unlike the rigid expert systems of the past, modern agents use LLMs to reason through problems dynamically. When faced with a task they haven't seen before, they don't crash; they analyse the goal, break it down into steps, and determine which software tools they need to use to solve it. This evolution has turned AI from a library of static knowledge into a dynamic engine of execution, capable of browsing the web, clicking buttons, and managing other software just as a human would.

The distinction between Generative AI and Agentic AI is best understood through the metaphor of an encyclopedia versus a skilled intern. Generative AI is like a magical encyclopedia: if you ask it a question, it gives you a brilliant, comprehensive answer, but it cannot do anything with that answer. It generates content but stops there. Agentic AI, on the other hand, functions like a smart intern. If you tell an agent, "Plan a marketing campaign," it doesn't just write the plan. It might proactively research the market, draft emails, log into the mailing software, schedule the messages, and then monitor the results. While Generative AI is reactive waiting for input to produce output Agentic AI is proactive. It operates in loops of perception and action: it assesses the environment, makes a plan, executes a step, checks the result, and corrects itself if something goes wrong.

This capability is powered by a cognitive cycle often described as the "OODA loop" Observe, Orient, Decide, Act. When you give an agent a goal, it first perceives its digital environment, erhaps by reading your emails or scanning a database. It then reasons about what needs

to be done, formulating a multi-step plan. Crucially, it then acts using "tools" software interfaces that allow it to browse the web, send messages, or run code. Finally, it reflects on the outcome. If an agent tries to book a flight and the website crashes, it does not just give up; it notices the error, interprets it, and tries a different airline or website. This ability to self-correct and persist in the face of obstacles is the defining characteristic of agency.

Real-world applications of this technology are already reshaping personal and professional tasks, with travel planning serving as a prime example of the agentic difference. In the previous era of simple chatbots, asking an AI to "book a trip to London" would result in a text itinerary that the user would have to manually book, visiting airline and hotel websites one by one. An Agentic AI, however, acts as a comprehensive concierge. It connects to flight APIs to check real-time availability, interacts with hotel booking engines to find rooms that match specific preferences such as “quiet” or “gym access,” and orchestrates ground transport logistics. If a flight is delayed, the agent perceives this change and autonomously rebooks the airport transfer and notifies the hotel of the late arrival, managing the entire sequence of events without user intervention.

In the corporate world, these agents are transforming into specialised digital workers within sectors like software development and supply chain management. We are moving away from the “copilot” model, in which a human types code and an AI suggests completions, towards fully autonomous coding agents. These agents can be assigned a high-level task, such as "refactor this database integration." The agent then explores the codebase, identifies the necessary files, writes new code, runs the test suite to ensure nothing is broken, and even fixes its own errors if the tests fail. Similarly, in supply chains, agents act as autonomous watchdogs that can detect a weather disruption in one

part of the world and automatically reroute shipments or order stock from an alternative supplier to prevent delays, notifying human managers only when high-level approval is required.

Ultimately, the rise of Agentic AI signals a fundamental shift in how humans interact with technology. We are transitioning from being operators of software to being managers of digital workforces. This brings immense potential for productivity, as one person can now orchestrate the output of many agents, but it also necessitates a new focus on governance and "human-in-the-loop" oversight. As these agents become more capable of taking real-world actions, the challenge shifts from prompt engineering to defining clear goals and safety establishing guardrails. We are no longer just chatting with machines; we are trusting them to act on our behalf, marking the beginning of a truly collaborative era between human intent and machine execution.

Mr. T. Thayatharsan
Graduand,
Department of Computer Science,
University of Jaffna.

Technical Mastery:
Front-End Development Series

The Computer Society (CompSoc) of the University of Jaffna organized a comprehensive training program titled “From Student to Professional: Front-End Development Series” to equip undergraduates with the skills to create and manage websites using modern web technologies. The series commenced on 24 June 2025 and concluded on 26 August 2025, comprising seven sessions—conducted both online and in-person.

The first two sessions were held via Zoom from 7.00 p.m. to 9.00 p.m., while the remaining sessions were conducted in-person from 1.00 p.m. to 3.00 p.m. at the CSL 3 & 4 Labs. The sessions were conducted by **Mr. Parathan Thiyagalingam**, **External Consultant – Full Stack Web Developer.**

A total of 122 undergraduates registered for the series, with active participation recorded across all sessions. The series covered essential topics including JavaScript fundamentals, variables, loops, arrays, DOM manipulation, async/await, promises, frameworks and libraries such as React, React + Vite setup, component basics, useState hook, React Router, GitHub version control, and deployment using Vercel.

Each in-person session included a recap, live coding demonstrations, debugging, and quizzes to assess understanding. Certificates of Participation were awarded to 34 students who maintained over 80% attendance, recognizing their commitment and engagement throughout the series.

The program concluded successfully, marking a transformative learning journey where undergraduates gained practical front-end development knowledge and confidence to build modern web applications, preparing them for professional growth and future opportunities.

STUDENT ACHIEVEMENTS

1. Appointed AWS Cloud Club Captain at University of Jaffna

Pamuda U. de A. Goonatilake, a Level 3 student of the Department of Computer Science, has been appointed as the AWS Cloud Club Captain for the University of Jaffna by Amazon Web Services. This appointment recognizes his leadership and dedication to promoting cloud computing skills among students.

2. National ICT NBQSA Gold Award 2025

Mr. Nimantha Rathnayake, a Graduate student of the Department of Computer Science, University of Jaffna, has brought pride to the university by winning the Gold Award under the Tertiary Student Project (Technology) category at the prestigious National ICT NBQSA Awards 2025.

His award-winning research, titled "A Deep Learning-Based Automated Tool for Ki-67 Quantification in Breast Cancer Histopathology," demonstrates an outstanding blend of technology and medical science, offering innovation in cancer diagnostics.The project was academically supervised by Prof. A. Ramanan, co-supervised by Dr. T. Kokul of the University of Jaffna, and medically supervised by Dr. Prathibha Nellihela with co-supervision by Dr. (Mrs.) Jayanjana Asanthi of the Teaching Hospital, Anuradhapura.

The prestigious award was presented at the Grand Awards Ceremony organised by the BCS Sri Lanka Section, endorsed by the Ministry of Digital Economy, held on 10 October 2025 at Hotel Taj Samudra, Colombo.

3. Team CodeRangers Crowned Champions at HackForce'25 with "UniConnect" Innovation

Four of our Level 2 students demonstrated outstanding innovation at HackForce'25, the Salesforce Agentforce Hackathon held under the theme Agentforce Innovation. The event gathered top student innovators from across the country to develop creative solutions using Salesforce technologies. The team CodeRangers impressed the judges with their project UniConnect and secured 1st place, becoming the champions of HackForce'25. UniConnect is a smart academic management system that brings together all essential academic information—such as course details, timetables, assignment deadlines, and exam schedules—into a single platform. It also allows students to easily book appointments with lecturers, improving communication and saving time for both students and staff.

Team Members

- Mr. Pasidu Chamod (2022/CSC/054) – Team Leader
- Mr. Pamuda U de A Goonatilake (2022/CSC/093)
- Mr. Himath Hansana Senasinghe (2022/CSC/068)
- Mr. Vihanga Nethsara (2022/CSC/042)

4. Team Why Not Us Secures First Runner-Up at Codestorm AI – YGC Innovation Festival 2025

Ms. Varsha Jeyarajalingam (2021/CSC/080), Mr. Tharmaraj Thayatharsan (2020/CSC/059), and Mr. Sanjay Kirupaharan (2022/CSC/066) showcased their AI expertise at Codestorm AI, a competition held as part of the YGC Innovation Festival 2025, organised by the Yarl IT Hub on 30th August 2025. The team delivered an outstanding performance and was awarded the First Runner-Up position, receiving a cash prize of LKR 30,000 in recognition of their achievement.

5. Team Thamizhi Wins at iCIIT Conclave 2025 Tamil LLMs Category

As part of the iCIIT Conclave's shared task organised by Informatics Institute of Technology (IIT), Sri Lanka on "Building Compact Sinhala & Tamil LLMs", held in August 2025, the team "Thamizhi" developed a compact Tamil language model designed for local language applications. At the final event on 5th August 2025, the team won under the category of Tamil LLMs and was recognised with certificates and prizes for their outstanding achievement. This work was carried out by Ms. Varsha Jeyarajalingam (2021/CSC/080), Mr. Tharmaraj Thayatharsan (2020/CSC/059), and Mr. Sanjay Kirupaharan (2022/CSC/066), under the supervision of Dr. Kengatharaiyer Sarveswaran.

6. SLASSCOM National Ingenuity Awards 2025 (Best innovative project – University Category)

The research project titled "A Deep Learning-Based Automated Tool for Ki-67 Quantification in Breast Cancer Histopathology: A Sri Lankan Context", conducted by Mr. Nimantha Rathnayake, a Graduate student of the Department of Computer Science, under the academic supervision of Prof. A. Ramanan, co-supervision of Dr. T. Kokul, and medical supervision of Dr. Prathibha Nellihela, co-supervision of Dr. (Mrs.) Jayanjana Asanthi, and Dr. Vathsal Bandaranayake of the Teaching Hospital, Anuradhapura, has been awarded "Best Innovative Project – University Category" at the provincial level of the Sri Lanka Association of Software and Services Companies (SLASSCOM) National Ingenuity Awards 2025. The award was presented to Mr. Nimantha Rathnayake at the Grand Awards Ceremony organised by SLASSCOM on the 1st of July 2025 at the Taj Samudra, Colombo.

SL-SWCS 2025

The eighth national workshop, **Sri Lanka Student Workshop on Computer Science (SL-SWCS'25)**, was held on 13 December 2025, organized by the Department of Computer Science, with approximately 150 participants attending physically.

Workshop on
ICT, AI & Cybersecurity for School Students

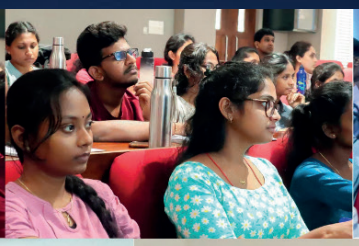
The Computer Society (CompSoc) successfully conducted its School Outreach Programme 2025 on 31 May 2025, introducing school students to key concepts in Information and Communication Technology (ICT), Problem-Solving, Artificial Intelligence (AI), and Cybersecurity. The programme was attended by 80 students from grades 6 to 11, representing the Vavuniya North Education Zone.

The sessions covered a range of topics, including an introduction to ICT and the IT field, where students learned about basic computing concepts, the internet, and the role of technology in various professions. In the problem-solving component, students explored structured techniques for addressing real-world challenges and were introduced to how generative AI can support the problem-solving process. The fundamentals of AI and its applications across industries were also discussed, alongside demonstrations of AI tools such as ChatGPT, Gemini, Leonardo, and Runway ML for educational and creative purposes. In addition, students were taught the essential principles of cybersecurity to better understand online safety and the protection of digital assets.

The programme adopted a hands-on approach, where students were asked to identify real-world problems and propose solutions using the knowledge gained during the sessions. The event concluded with student presentations of their proposed solutions, and the top-performing team was recognised and awarded prizes based on creativity, practicality, and presentation quality.

The sessions were delivered by members of CompSoc, with project evaluations carried out by Professor A. Ramanan, Dr. Samantha Tharani Jeyakumar, and Assistant Lecturers Ms. T. Kayalvizhy, and Ms. S. Sweshthika. We extend our appreciation to the WIE, IEEE Student Branch Affinity Group of the University of Jaffna for their logistical support in organising and delivering the event.

CompSoc
EVENT GALLERY



FROM STUDENT TO PROFESSIONAL:
CAREER FAIR PREP ESSENTIALS
Guest Speaker
Christopher Wijesinghe
Talent Evangelist
Calsoy Technologies
Guest Speaker
Chomal Miguntenne
Senior Tech Lead
Calsoy Technologies
Date
04 March 2025
Time
1 pm - 3 pm
Venue
CSA , DCS , UOJ

COMPUTER SOCIETY
PRESENTS
T-SHIRT
DESIGN CONTEST
CREATE YOUR DESIGN
WINNER GETS A FREE T-SHIRT
DEADLINE
06-APRIL-2025
DEADLINE EXTEND
FOR MORE DETAILS
071-641-6709

COMPUTER SOCIETY
PRESENTS
WINNING
DESIGN
Congratulations
MR. S. M. SAHAM
LEVEL 2, DEPT. OF COMPUTER SCIENCE
WINNER OF
T-SHIRT DESIGN CONTEST

PRESENTS
COMING SOON

FINAL ROUND
12 HOURS
INTER-UNIVERSITY
CODING COMPETITION
3:30 ONWARDS @ DCS, UOJ
WINNER
LKR. 50,000.00
1ST RUNNER-UP
LKR. 30,000.00
2ND RUNNER-UP
LKR. 20,000.00
Platinum Sponsor
apptimus
Silver Sponsor
Unicom SD

ELIMINATION ROUND
TODAY
6.30P.M. - 8.00P.M.

LANGUAGE TECHNOLOGY WORKSHOP
Topics
Join us to learn how computers learn structure of the languages !
Open only to registered participants !
Date
16th of December 2025
Time
9:00 A.M. - 2:00 P.M.
Venue
CSL 3 & 4, DCS, UOJ

DIGITIZATION OF HISTORICAL DOCUMNETS
Session Highlights
Guest Speaker
Dr. Durrat ul Ain Akram
Assistant professor
IX Department, UET
Consultant CLE, KICS, UET
23rd December 2025
1.15 p.m. - 2.15 p.m.
Auditorium

Career Guidance Unit
University of Jaffna
Career Guidence Cell
Faculty of Science
From Code to Strategy :
My Journey from IBM Principal to Chief AI Officer
Guest Speaker
Arjuna Swaminathan
A visionary Chief AI Officer & Former IBM Principal
Key Takeaways
• Discuss different AI career paths.
• Learn essential technical skills.
• Understand the leadership mindset needed for innovation.
6th of January, 2026
1 PM - 3 PM
Auditorium, Department of Computer Science

Career Guidance Unit
University of Jaffna
Career Guidence Cell
Faculty of Science
The Entrepreneurial Mindset :
Human Skills for a High-Tech World
Guest Speaker
Arjuna Swaminathan
A visionary Chief AI Officer & Former IBM Principal
Key Takeaways
• Understand the role of human skills in an AI-driven world.
• Learn practical communication for professional success.
• Build confidence, empathy, and resilience for a global career.
7th of January, 2026
1 PM - 3 PM
Auditorium, Department of Computer Science

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University of Jaffna
2022/2023

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